Department of Chemical Engineering, Chulalongkorn University

Teaching document · operator training simulator

Cyclohexane Reactor Ver 1

HAZOP STUDY REPORT

Benzene hydrogenation to cyclohexane: feed, R-101 / R-102 slurry reactors, R-103 finishing bed, hydrogen loop, steam and utilities

Document no.	CHX-V1-SA-HZP-001
Revision	A · 2026-10-26 · Issued for teaching
Unit	Cyclohexane Reactor Ver 1
Client	Department of Chemical Engineering, Chulalongkorn University
Method	IEC 61882:2016 hazard and operability study, example 5 × 5 risk matrix
Drawings	CHX-V1-PID-001 (screen P&ID) · CHX-V1-PFD-001 (PFD) · CHX-V1-IN-CE-001 (C&E)

This is a teaching HAZOP of a training simulator, not the HAZOP of a real plant. The safeguards are the alarms, trips, permissives and fail positions that exist in the simulator; every limit and design condition is read from the simulator model. Names, signatures and dates of the team are left blank on purpose.

Revisions

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
A	2026-10-26	Issued for teaching	Apinan Soottitantawat		

HAZOP close-out summary

One line per recommendation (38 actions). "Raised from" lists the worksheet rows; the remark gives the highest risk the action addresses. Action taken is filled in by the action party; status stays OPEN until the HAZOP leader accepts the response (Appendix 3).

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 N7	**R01**	N1.2 N7.1	Add a high-high benzene flow or benzene-to-hydrogen ratio trip (FI-101 against FI-103) that closes XV-101; the feed can run far above design with no trip.	Instrument / Process		OPEN	Highest risk 12 · High
N2 N4 N8	**R02**	N2.1 N2.7 N4.5 N8.4	Confirm and document the pressure relief of R-101, R-102, R-103 and V-104 for blocked outlet, make-up valve wide open and fire; relief valves are not shown on the training P&ID.	Process		OPEN	Highest risk 12 · High
N1 N2 N4 N5	**R03**	N1.2 N2.4 N4.1 N5.1	Add a high-high temperature trip on R-102 with its own element (set below the R-102 boiling point at loop pressure).	Instrument / Safety		OPEN	Highest risk 12 · High
N2	**R04**	N2.4	Add low-low agitator power (JI-101) to the SIS so that a running motor with a failed drive is seen like a stopped motor.	Instrument / Safety		OPEN	Highest risk 12 · High
N2	**R05**	N2.5 N2.6	Add an independent R-101 level element with high-high and low-low actions (stop feed; protect P-101A/B).	Instrument		OPEN	Highest risk 9 · Medium
N4	**R06**	N4.1 N4.4	Add low cooling-water flow on E-103 or high-high TI-103 to stop the benzene feed.	Process / Instrument		OPEN	Highest risk 12 · High
N5	**R07**	N5.3 N5.4	Add independent high-high and low-low level on R-102 (stop P-102A/B, stop feed).	Instrument		OPEN	Highest risk 9 · Medium
N1	**R08**	N1.4	Fit a non-return valve on the P-100A/B discharge and close XV-101 when no feed pump runs.	Process / Mechanical		OPEN	Highest risk 10 · High
N1	**R09**	N1.7 N1.8	Analyse every benzene receipt for sulphur and water; drain TK-100 water; consider a guard bed ahead of R-101.	Process / Operations		OPEN	Highest risk 9 · Medium
N1 N3 N6	**R10**	N1.1 N3.1 N6.1	Auto-start the standby pumps P-100B, P-101B and P-102B on low discharge pressure or flow.	Electrical / Instrument		OPEN	Highest risk 8 · Medium
N3	**R11**	N3.1	Add a low-flow alarm on the pump-around FI-104 (none in the alarm list).	Instrument		OPEN	Highest risk 8 · Medium
N3	**R12**	N3.3	Add a control-margin alarm on the FIC-104 output and trend the E-101 heat-transfer coefficient.	Process / Instrument		OPEN	Highest risk 9 · Medium
N3	**R13**	N3.4	Confirm the D-101 steam-side relief for blocked export at maximum E-101 duty.	Process		OPEN	Highest risk 9 · Medium
N3	**R14**	N3.6 N3.7	Add low-low and high-high level on D-101 independent of LIC-109.	Instrument		OPEN	Highest risk 9 · Medium
N3	**R15**	N3.8	Monitor D-101 blowdown or condensate for hydrocarbon (E-101 tube leak).	Process		OPEN	Highest risk 8 · Medium
N3 N10	**R16**	N3.9 N10.4	Interlock HV-110 (MP steam to E-101) closed whenever XV-101 is open, and put the HEAT to COOL switch-over in the light-off steps.	Instrument / Operations		OPEN	Highest risk 12 · High
N4 N5	**R17**	N4.6 N5.6	Detect hydrogen and hydrocarbon in the CW return of E-103, E-106 and the R-102 jacket (degassing pot or analyser).	Process		OPEN	Highest risk 8 · Medium
N2 N5 N7 N10	**R18**	N2.10 N5.2 N7.7 N10.9	Route product to TK-107 automatically on AI-103 or AI-104 high and alarm a stale analyser value.	Instrument / Operations		OPEN	Highest risk 6 · Medium

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N7	**R19**	N7.6	Add an independent low-low level on V-104 closing a shutdown valve in the product line; check the TK-106 / TK-107 vents for gas blow-by.	Process / Instrument		OPEN	Highest risk 10 · High
N6	**R20**	N6.4	Add a low ΔP alarm on PDI-102 (torn candle) and a routine fines check of the product.	Instrument / Operations		OPEN	Highest risk 6 · Medium
N2 N4 N8	**R21**	N2.7 N4.5 N8.4	Add a PI-101 high alarm between the PIC-105 alarm and PSHH-101, because PIC-105 measures at V-104, not at R-101.	Instrument		OPEN	Highest risk 10 · High
N7	**R22**	N7.3	Add a high alarm on TI-109 (V-104 temperature).	Instrument		OPEN	Highest risk 9 · Medium
N8	**R23**	N8.7	Add a low-flow alarm on the purge FI-107 and a purity check in the shift routine.	Instrument / Operations		OPEN	Highest risk 9 · Medium
N8	**R24**	N8.2	Add low-low hydrogen flow (FI-103) to the SIS to close XV-101; today only a stopped K-101 motor is seen.	Instrument / Safety		OPEN	Highest risk 12 · High
N8	**R25**	N8.6	Install hydrogen gas detection at K-101, the reactors and V-104, and alarm the loop hydrogen balance (make-up against 3 × benzene fed).	Safety / Instrument		OPEN	Highest risk 10 · High
N8 N10	**R26**	N8.8 N10.1	Add a second loop oxygen analyser (2oo2) for the XV-105 permissive and record the O2 readings in the purge procedure.	Instrument / Operations		OPEN	Highest risk 10 · High
N8	**R27**	N8.9 N8.10	Fit a non-return valve on the K-101 discharge and confirm the kickback / anti-surge protection and its high-temperature alarm.	Mechanical		OPEN	Highest risk 9 · Medium
N9	**R28**	N9.2	Protect the nitrogen connection: non-return valve, double block and bleed or a removable spool; HV-120 opened only below the header pressure.	Process / Mechanical		OPEN	Highest risk 12 · High
N9	**R29**	N9.3 N9.4 N9.7 N9.8	Depressuring study for EDV-101 (rate, flare load, minimum metal temperature); fail-open actuator and proof test.	Process		OPEN	Highest risk 10 · High
N9	**R30**	N9.5 N9.6	Provide a flare knock-out drum and a continuous flare purge or seal to keep air out.	Process		OPEN	Highest risk 8 · Medium
N1 N2 N6 N10	**R31**	N1.7 N2.9 N6.7 N10.5 N10.6	Write a Raney-Ni handling procedure: always wet and under nitrogen, metered charges through HV-117, filter-cake handling.	Operations / Safety		OPEN	Highest risk 12 · High
N2 N3 N10	**R32**	N2.2 N3.2 N10.3 N10.8	Light-off and shutdown steps: limit the feed ramp after XV-101 opens, stop benzene before hydrogen, and add a low-temperature-with-feed interlock (TI-100 with XV-101 open) to stop benzene accumulation.	Operations / Instrument		OPEN	Highest risk 12 · High
N2	**R33**	N2.1	Add a TIC-101 against TI-100 deviation alarm (two elements in the same reactor).	Instrument		OPEN	Highest risk 12 · High
N6 N10	**R34**	N6.2 N6.7 N10.7 N10.10	Gas-free test and positive isolation before opening any equipment (filters, reactors); written filter change-over procedure.	Operations		OPEN	Highest risk 12 · High
N1 N6	**R35**	N1.1 N1.5 N6.5	Fit minimum-flow bypasses on P-100A/B and P-102A/B.	Mechanical		OPEN	Highest risk 9 · Medium
N7	**R36**	N7.8	Review the blanketing and venting of TK-106 / TK-107 for hydrogen flashing from the product.	Process		OPEN	Highest risk 9 · Medium
N1 N7	**R37**	N1.2 N7.1 N7.9	Add an R-103 bed-rise alarm (TI-107E minus TI-106) as early warning of benzene slip.	Instrument		OPEN	Highest risk 12 · High
N5	**R38**	N5.5	Positive segregation of MP steam and CW on the R-102 jacket (interlocked change-over) and jacket relief.	Process / Mechanical		OPEN	Highest risk 6 · Medium

Cyclohexane Reactor Ver 1

HAZOP study report

Document CHX-V1-SA-HZP-001 · Rev A · 2026-10-26

Approvals

Roles only: the names, signatures and dates are completed by the people who hold these roles when the study is run.

Action	Role	Name	Signature	Date
Prepared by	HAZOP scribe / study secretary			
Reviewed by	HAZOP leader (independent facilitator)			
Reviewed by	Process engineer			
Reviewed by	Operations representative			
Reviewed by	Instrument / control engineer			
Reviewed by	Safety / loss prevention			
Approved by	Unit owner / plant manager			

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Executive summary

The team examined **10 nodes** of the Cyclohexane Reactor Ver 1 unit with eleven guide words and recorded **87 deviations** (full recording). After credit for the existing safeguards the risk is **12 Low, 55 Medium, 20 High and 0 Very high**. The study raised **38 recommendations** (R01–R38); every High row has at least one. 9 of the 20 High rows rely on a trip and go to LOPA.

Risk level	R = S × L	Rows	Share	Action
Low	1–4	12	14 %	Accept; manage by procedures and training
Medium	5–9	55	63 %	Reduce as low as reasonably practicable; recommendations reviewed by the team
High	10–16	20	23 %	Not acceptable long-term: recommendation mandatory, LOPA before start-up
Very high	20–25	0	0 %	Intolerable: redesign or add independent protection before operation

Key findings

- **More benzene has no trip of its own.** With FIC-101 driven to 100 % the R-103 bed reached 208.8 °C (TSHH-107 at 220 °C) and R-102 206.5 °C in 30 min with no trip; the protection was 8 alarms and a product at 23,728 ppm benzene (R01, R03).
- **Two failures the SIS cannot see.** A drive that fails while the motor runs (D4) and a hydrogen valve that closes while K-101 runs (D9) gave no trip in 30 min: R-101 fell to 70 °C and 112 °C with benzene building up, and in D4 R-102 reached 221.6 °C, which has no trip at all. XS-101 and XS-102 watch the motors, not the duty (R03, R04, R24).
- **Loss of pump-around is caught, just.** TSHH-100 tripped at 02:57 after the first alarm (TIC-101 HI at 00:57); TI-100 peaked at 220.3 °C, about two minutes for the operator to act before the SIS does (R11).
- **Losing the reflux condenser does not trip the unit.** TI-103 alarmed at 00:18, R-102 rose to 196.9 °C and R-101 level fell to 36.6 %, below the 40 % start permissive, with no trip in 30 min (R06).
- **A transmitter error moves the reactor close to its trip.** With the TIC-101 element reading low, R-101 settled at 215.0 °C, 5.0 °C under TSHH-100, and only the independent TI-100 alarm (02:13) showed it; isomer rose to 175 ppm against the 200 ppm spec (R33).
- **The existing trips work where they look.** Make-up wide open: PSHH-101 at 00:57 with PI-101 peaking at 33.58 bar g (set 33.5); K-101 motor stop: XS-102 at 00:11; V-104 filling: LSHH-104 stopped K-101 at 14:25 and XS-102 tripped at 14:35. A trip only limits the deviation it measures; relief is still needed for faster events (R02).
- **R-102 can fill with one alarm.** With LV-102 shut, R-102 reached 110 % (top of range) after a single LIC-102 HI alarm at 12:35 and no trip (R07).
- **Relief is not on the training P&ID.** Recommendation R02 asks the team to confirm the relief of R-101, R-102, R-103 and V-104; a trip only limits the deviation it measures.

Actions by party

Action party (discipline)	Actions	Recommendations
Instrument	23	R01 R03 R04 R05 R06 R07 R10 R11 R12 R14 R16 R18 R19 R20 R21 R22 R23 R24 R25 R26 R32 R33 R37
Process	15	R01 R02 R06 R08 R09 R12 R13 R15 R17 R19 R28 R29 R30 R36 R38
Operations	9	R09 R16 R18 R20 R23 R26 R31 R32 R34
Safety	5	R03 R04 R24 R25 R31
Mechanical	5	R08 R27 R28 R35 R38
Electrical	1	R10

A recommendation shared by two disciplines is counted under both (58 assignments for 38 actions).

1 Introduction

1.1 Purpose

This report records a hazard and operability (HAZOP) study of the cyclohexane unit modelled in the operator training simulator. It is a teaching document: it shows how a HAZOP is defined, run, recorded and followed up, using a unit whose every safeguard can be tried on the simulator.

1.2 The unit

Benzene is hydrogenated to cyclohexane (C6H6 + 3 H2 → C6H12, ΔH −206 kJ/mol). At design the simulator feeds 34.8 t/h of benzene to the stirred Raney-Ni slurry reactor R-101 at 200 °C and 30.0 bar g, which converts about 98 % and releases about 25.0 MW (E-101 removes 9.6 MW raising LP steam in D-101, the rest leaves as boil-up condensed in E-103). The liquid boils at about 258 °C at reactor pressure. R-102 (160 °C, jacketed) and the adiabatic trickle bed R-103 finish the conversion; the product is cooled in E-106 and separated in V-104 at 29.0 bar g. Hydrogen is recycled by K-101 (46 000 Nm³/h) with make-up under PIC-105 and a purge.

The safety instrumented system trips the unit on R-101 temperature (220 °C), R-103 bed temperature (220 °C), R-101 pressure (33.5 bar g), agitator or compressor stopped with feed, and the ESD pushbutton; V-104 level high-high (90 %) stops K-101. The full cause-and-effect matrix is document CHX-V1-IN-CE-001.

1.3 References

Reference	Title / body	How it is used here
IEC 61882:2016	Hazard and operability studies (HAZOP studies): application guide	The method used here: nodes, design intent, guide words, full recording, follow-up
CCPS, Guidelines for Hazard Evaluation Procedures	Center for Chemical Process Safety (AIChE)	Worksheet layout, team roles, risk ranking practice
IEC 61511	Functional safety: safety instrumented systems for the process industry	High rows that rely on a trip go to LOPA and SIL assessment (Safety layers, LOPA tabs)
ISO 31000	Risk management: guidelines	The risk matrix as one tool inside a managed risk process
Thai DIW regulation on hazard identification and risk assessment, B.E. 2543 (2000)	Department of Industrial Works, Ministry of Industry	Factories handling hazardous materials must identify hazards, assess risk and keep a risk management plan; HAZOP is an accepted method
API 520 / API 521	Pressure-relieving and depressuring systems	Relief and depressuring cases raised in nodes N2, N8 and N9
OSHA 29 CFR 1910.119 (where applicable)	Process safety management of highly hazardous chemicals (USA)	Process hazard analysis revalidated at least every five years; used as reference practice

2 Scope

2.1 Boundaries

Included: benzene from the TK-100 outlet through P-100A/B, FV-101 and XV-101; R-101 with its agitator, pump-around, E-101 and steam drum D-101; E-103 / V-103 reflux; R-102 with its jacket, P-102A/B and the catalyst filters F-102A/B; R-103, E-106, V-104 and the product line to TK-106 / TK-107 (HS-115); the hydrogen loop (K-101, FV-103, make-up PV-105 / XV-105, purge HIC-107); nitrogen HV-120, vent HV-121 and EDV-101 to the flare header; start-up, shutdown and catalyst procedures.

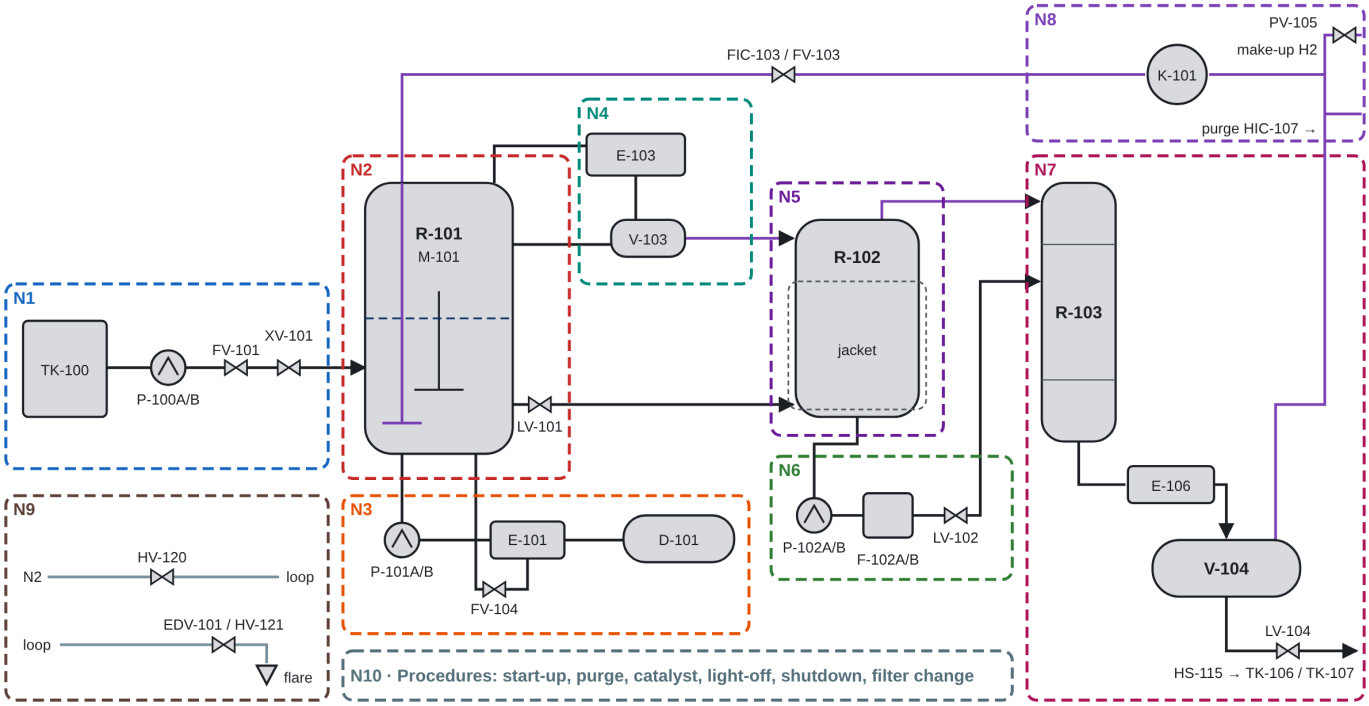
Excluded: the tank farm beyond TK-100 / TK-106 / TK-107, utility generation (cooling water, steam, nitrogen, hydrogen supply), the flare system beyond the unit header, electrical and control-room buildings. Utility failures are covered as causes inside the nodes.

2.2 Nodes

Node	Name	Design intent	Equipment	Rows
N1	Benzene feed	Deliver 34.8 t/h of benzene at 40 °C from TK-100 by P-100A/B under FIC-101 control through FV-101 and the SIS valve XV-101 into R-101 at 30.0 bar g.	TK-100, P-100A/B (one running, one standby), FV-101, XV-101	8
N2	R-101 slurry reactor	Hydrogenate about 98 % of the benzene in a stirred Raney-Ni slurry at 200.0 °C and 30.0 bar g with 65.0 % level, agitator M-101 dispersing the hydrogen from the sparger; liquid overflows by LIC-101 / LV-101 to R-102.	R-101, agitator M-101, hydrogen sparger, catalyst pot HV-117, LV-101	10
N3	Pump-around, E-101 and steam drum	Circulate 357.3 t/h of reactor slurry by P-101A/B through E-101 (FIC-104 in cascade from TIC-101, FV-104 fails open) and remove 9.6 MW by raising LP steam in D-101 at 4.50 bar g; HS-101 HEAT puts MP steam (HV-110) on E-101 for start-up only.	P-101A/B, E-101, FV-104, D-101, LV-109, PV-110, HV-110, HS-101	9

Node	Name	Design intent	Equipment	Rows
N4	Reflux condenser E-103 / V-103	Condense the cyclohexane boiled off R-101 with the hydrogen in E-103 on cooling water (HV-113), return 46.5 t/h of reflux from V-103 to R-101 and pass the gas at 55.7 °C to R-102.	E-103, V-103, HV-113, vent line V-103 → R-102	7
N5	R-102 and its jacket	Hydrogenate most of the remaining benzene at 160.0 °C and 29.6 bar g, with the jacket on MP steam (HS-102 HEAT, TIC-102 → TV-102) or cooling water (COOL); hold 55.0 % level.	R-102, jacket, TV-102, HS-102	7
N6	Transfer and catalyst filter	Pump 20.9 t/h of R-102 liquid by P-102A/B through the candle filter F-102A or B (PDI-102 0.26 bar clean) and LV-102 to R-103, keeping the Raney-Ni in R-102.	P-102A/B, F-102A/B, LV-102	7
N7	R-103, E-106, V-104 and product	Finish the benzene to about 2.00 ppm in the adiabatic trickle bed R-103 (158.6 °C outlet), cool in E-106, separate gas and liquid in V-104 at 29.0 bar g and 44.9 °C, and send 37.5 t/h of product by LIC-104 / LV-104 to TK-106 (on spec) or TK-107 (slop) by HS-115.	R-103, E-106, V-104, LV-104, HS-115, TK-106, TK-107	9
N8	Hydrogen loop	Recycle 46,000 Nm³/h of hydrogen by K-101 (discharge 32.0 bar g) through FIC-103 / FV-103 to the R-101 sparger; make up 31,974 Nm³/h by PIC-105 / PV-105 holding V-104 at 29.0 bar g; purge 2,017 Nm³/h by HIC-107 to hold purity at 92.1 %.	K-101, FV-103, PV-105, XV-105, HIC-107, loop piping R-101 → R-102 → R-103 → V-104 → K-101	11
N9	Nitrogen, vent, EDV-101 and flare	Purge the loop with nitrogen (8 bar g) through HV-120 before hydrogen and before maintenance; vent through HV-121 and depressure through EDV-101 to the flare.	HV-120, HV-121, EDV-101, flare header	8
N10	Procedures	Start up (nitrogen purge to below 1 % O2, hydrogen admission, catalyst charge, heat-up on MP steam, light-off at TI-100 ≥ 140 °C and feed ramp), shut down, and change over the catalyst filter without hazard to people or equipment.	Whole unit	11

2.3 Node map



Node map from the HAZOP tab of the simulator (dashed boxes in the node colours). A larger copy is in Appendix 2.

3 Methodology

3.1 Standard and phases

The study follows IEC 61882:2016. Each node is described by its design intent; parameters combined with guide words give deviations; for each deviation the team records causes, consequences judged *without* credit for safeguards, the existing safeguards, the risk ranking (likelihood *with* the existing safeguards) and recommendations. Every deviation considered is written down (full recording).

Phase	What is done
1 Definition	Scope and boundary, objectives, team, the risk matrix to be used
2 Preparation	Up-to-date documents, nodes and their design intents, parameter and guide-word list, schedule
3 Examination	For each node, parameter and guide word: deviation, causes, consequences (without safeguards), existing safeguards, risk, recommendations
4 Documentation and follow-up	Worksheets, recommendation register, report, responses and close-out, revalidation

3.2 Guide words

Guide word	Meaning	Typical deviations	Rows
NO / NOT	Complete negation of the design intent	No flow, no mixing, no cooling water, a valve that does not open	11
MORE	Quantitative increase	More flow, higher pressure, temperature, level or activity	28
LESS	Quantitative decrease	Less flow, lower pressure, temperature, level, purity	18
AS WELL AS	Qualitative modification: something extra is present	Water, oxygen, catalyst fines or hydrocarbon where they should not be	9
PART OF	Qualitative modification: only part of the intent is achieved	A stream short of one component; a procedure with a step left out	2
REVERSE	Logical opposite of the intent	Back-flow through a stopped pump or compressor, into a lower-pressure system	5
OTHER THAN	Complete substitution: something else happens	Wrong material, wrong medium, maintenance instead of operation, a leak	7
EARLY	Relative to clock time: too early	A step done before the plant is ready for it	2
LATE	Relative to clock time: too late	A switch-over or response that comes after it was needed	2
BEFORE	Relative to order or sequence: ahead of its turn	Hydrogen before the oxygen is purged; benzene before light-off	2
AFTER	Relative to order or sequence: behind its turn	Catalyst charged after the feed is already at full rate	1

3.3 Parameters

Parameter	What it covers on this unit	Rows
Flow	Liquid and gas flows in each line, utility flows (CW, steam, N2), relief and vent flows	26
Pressure	Vessel, loop, pump and compressor discharge, utility side of exchangers	12
Temperature	Reactor, bed, separator, exchanger outlet, metal temperature	12
Level	R-101, R-102, V-104, D-101 inventories	7
Composition	Benzene, hydrogen purity, oxygen, sulphur, water, isomer, leaks into utilities	12
Reaction	Catalyst activity and amount in R-101 and R-103	2
Mixing	Agitation and gas dispersion in R-101	2
Utility	Which utility is on which exchanger (HS-101 / HS-102 selectors)	1
Containment	Leaks to atmosphere from the hydrogen loop	1
Maintenance	Opening equipment, filter change-over, catalyst handling	2
Sequence	Order of steps in start-up, light-off and shutdown	6
Time	Timing of steps and of responses to alarms	4

3.4 Risk matrix (example, 5 × 5)

The matrix is an example for teaching (typical practice); S and L tables below. Each cell shows R = S × L, coloured by band, and the number of worksheet rows that fell in it.

Severity S		L1 Remote	L2 Unlikely	L3 Possible	L4 Likely	L5 Frequent
S5	Catastrophic	**5** · 1 row	**10** · 7 rows	**15**	**20**	**25**
S4	Major	**4** · 2 rows	**8** · 10 rows	**12** · 13 rows	**16**	**20**
S3	Moderate	**3** · 1 row	**6** · 7 rows	**9** · 19 rows	**12**	**15**
S2	Minor	**2**	**4** · 4 rows	**6** · 16 rows	**8** · 2 rows	**10**
S1	Negligible	**1**	**2** · 1 row	**3** · 2 rows	**4** · 2 rows	**5**

S	Severity	People / asset / environment
1	Negligible	No injury; small off-spec batch; < 10 k$
2	Minor	First-aid case; short outage; contained spill; < 100 k$
3	Moderate	Lost-time injury; unit down for days; reportable release; < 1 M$
4	Major	Single fatality or permanent disability; major equipment damage; off-site effect; < 10 M$
5	Catastrophic	Multiple fatalities; loss of the unit; major off-site impact; > 10 M$

L	Likelihood	Frequency band
1	Remote	$< 10^{-4}$ per year: not expected in the life of the industry
2	Unlikely	10^{-4} to 10^{-3} per year: heard of in the industry
3	Possible	10^{-3} to 10^{-2} per year: has happened in the company
4	Likely	10^{-2} to 10^{-1} per year: once in the life of the unit
5	Frequent	$> 10^{-1}$ per year: several times in the life of the unit

4 Sessions

4.1 Schedule

Dates are left blank: they are set when the study is run. Session lengths are typical practice (assumed), about 15 to 20 worksheet rows per half-day.

Session	Nodes / topic	Rows	Date	Duration
0	Kick-off: scope, method, risk matrix, walk through the P&ID and the C&E	—		half day (assumed)
1	N1, N2	18		half day (assumed)
2	N3, N4	16		half day (assumed)
3	N5, N6	14		half day (assumed)
4	N7, N8	20		half day (assumed)
5	N9, N10 (procedures)	19		half day (assumed)
6	Review of the draft report and the recommendation register	—		half day (assumed)

4.2 Team

Roles as recommended by IEC 61882 and CCPS; names are left blank.

Role	Brings to the study	Name	Sessions attended
Leader (independent)	Runs the method, keeps the pace and the scope, not from the design team so assumptions get challenged.		
Scribe	Records every row in the team's words while the team watches the screen.		
Process engineer	Design intent, heat and mass balance, reaction and relief cases.		
Operations	How the unit is really run: start-up, shutdown, what goes wrong on shift, what operators can see and do.		
Instrument / control	Loops, fail positions, alarm limits, SIS logic and its independence.		
Safety / loss prevention	Consequence judgement, risk matrix, link to LOPA and SIL.		
Specialists as needed	Catalyst supplier (Raney-Ni handling), compressor, metallurgy, electrical.		

4.3 Documents reviewed

Document	Title	Where in the simulator
CHX-V1-PID-001	P&ID (screen P&ID P&ID-001, rev 0)	P&ID tab
CHX-V1-PFD-001	Process flow diagram	PFD tab and files/CHX-V1-PFD-001
CHX-V1-IN-CE-001	Cause and effect matrix of the SIS, alarms and permissives (rev R0)	C&E matrix tab
CHX-V1-IN-SRS-001	Safety requirements specification of the SIFs (reserved)	SRS of SIF tab
—	Process description and design basis	Process tab and Design of the unit operation tab (design point, duties, sizes)
—	Flow diagrams	BFD and PFD tabs (streams and their conditions)
—	P&ID	P&ID tab: every tag, valve and instrument named in the worksheets
—	Control philosophy	Control loop design tab: loops, cascades, fail positions
—	Alarm list, trip list and permissives	Safety system (SIS) tab and the alarm limits of the DCS console
—	The plant itself	The simulator: part 9 runs deviations on copies of it

4.4 Guide-word coverage

Number of worksheet rows per node and guide word ("·" = considered, nothing credible recorded).

Node	NO/NOT	MORE	LESS	AS WELL	PART OF	REVERSE	OTHER	EARLY	LATE	BEFORE	AFTER	Rows
N1	1	3	1	1	·	1	1	·	·	·	·	8
N2	2	3	4	1	·	·	·	·	·	·	·	10
N3	1	4	2	1	·	·	1	·	·	·	·	9
N4	1	3	1	1	·	1	·	·	·	·	·	7
N5	1	3	1	1	1	·	·	·	·	·	·	7
N6	1	3	1	1	·	·	1	·	·	·	·	7
N7	1	4	3	·	·	·	1	·	·	·	·	9
N8	1	4	3	1	·	1	1	·	·	·	·	11
N9	2	1	2	2	·	1	·	·	·	·	·	8
N10	·	·	·	·	1	1	2	2	2	2	1	11
All	11	28	18	9	2	5	7	2	2	2	1	87

5 Findings

5.1 Risk profile per node

Node	Name	Rows	Low	Medium	High	Very high	Highest	Recs
N1	Benzene feed	8	2	4	2	0	12 · High	8
N2	R-101 slurry reactor	10	1	5	4	0	12 · High	9
N3	Pump-around, E-101 and steam drum	9	0	9	0	0	9 · Medium	8
N4	Reflux condenser E-103 / V-103	7	2	4	1	0	12 · High	5
N5	R-102 and its jacket	7	0	6	1	0	12 · High	5
N6	Transfer and catalyst filter	7	2	4	1	0	12 · High	5
N7	R-103, E-106, V-104 and product	9	1	6	2	0	12 · High	6
N8	Hydrogen loop	11	1	6	4	0	12 · High	7
N9	Nitrogen, vent, EDV-101 and flare	8	1	5	2	0	12 · High	3
N10	Procedures	11	2	6	3	0	12 · High	6
All		87	12	55	20	0		38

5.2 High-ranked rows

20 rows rank High or Very high after safeguards; those relying on a trip are marked † (LOPA candidates).

Row	Deviation	Main consequence	Safeguards	S	L	R	Recs
N1.2 †	Flow MORE: More benzene than the hydrogen and heat removal can take	Conversion in R-101 falls and unreacted benzene passes to R-102 and R-103, which then release its heat of reaction: R-102 and the adiabatic R-103 bed heat up (about 9 °C per 1 % benzene slip), the product goes off-spec and the bed can run away.	FIC-101 HI alarm at 42 t/h; TIC-102 HI alarm at 172 °C; TI-107 HI alarm at 185 °C; TSHH-107 trips at 220 °C (5 s); TSHH-100 trips at 220 °C (5 s); AI-103 HI alarm at 10 ppm	4	3	12	R01 R03 R37
N1.4	Flow REVERSE: Reactor liquid and hydrogen flow back toward P-100A/B and TK-100	Hot liquid saturated with hydrogen enters the feed line and the atmospheric tank TK-100: tank overpressure, release of hydrogen and benzene vapour, fire.	XV-101 closes on any SIS trip; FV-101 fails closed; FIC-101 LO alarm at 25 t/h	5	2	10	R08
N2.1 †	Temperature MORE: R-101 above 200.0 °C toward its boiling point (about 258 °C at reactor pressure)	Faster reaction and more isomer (MCP); at the boiling ceiling the extra heat becomes vapour, the pressure and the condenser load rise and hot hydrocarbon is carried to R-102.	TIC-101 HI alarm at 208 °C; TI-100 independent element HI alarm at 210 °C; TSHH-100 trips at 220 °C (5 s): XV-101 and XV-105 shut, K-101 stops, E-101 to COOL with FV-104 fully open; PSHH-101 trips at 33.5 bar g	4	3	12	R02 R33
N2.2	Temperature LESS: R-101 below the light-off temperature with feed in	The reaction slows while benzene keeps coming: unreacted benzene accumulates in R-101 and slips to R-102.	TIC-101 LO alarm at 190 °C; XV-101 alarm when benzene is permitted with R-101 below 180 °C; feed permissive TI-100 ≥ 140 °C (start only)	4	3	12	R32
N2.4	Mixing LESS: Impeller turns slowly while the motor runs	As for no agitation, but the motor-run signal stays healthy so XS-101 does not act: benzene builds up in R-101 and R-102 receives it; R-102 has no high-high temperature trip.	JI-101 LO alarm at 80 kW; TIC-101 LO alarm at 190 °C; TIC-102 HI alarm at 172 °C	4	3	12	R04 R03
N2.7 †	Pressure MORE: R-101 pressure above normal	Overpressure of R-101 and the loop: loss of containment of hydrogen and hot hydrocarbon, fire or explosion.	PIC-105 HI alarm at 30.5 bar g (measured at V-104); PSHH-101 trips at 33.5 bar g (3 s): XV-105 make-up and XV-101 shut, K-101 stops	5	2	10	R02 R21
N4.1 †	Flow NO / NOT: No cooling water to E-103	Reflux FI-105 stops: R-101 loses a heat sink and its level falls; hot cyclohexane vapour goes with the hydrogen to R-102, which heats, and on to R-103.	TI-103 HI alarm at 75 °C; LIC-101 LO alarm at 45 %; TIC-102 HI alarm at 172 °C; TSHH-107 trips at 220 °C; feed permissive HV-113 open (start only)	4	3	12	R06 R03
N5.1 †	Temperature MORE: R-102 hotter than 160.0 °C	More vapour and benzene go on to R-103, whose adiabatic bed heats; R-102 approaches its boiling ceiling and has no high-high trip of its own.	TIC-102 HI alarm at 172 °C; TV-102 fails closed (no steam in HEAT); any SIS trip sets HS-102 to COOL; TSHH-107 trips at 220 °C	4	3	12	R03
N6.7	Maintenance OTHER THAN: Filter opened for cleaning while not isolated or not gas-free	Hydrogen and hot hydrocarbon released when the filter is opened; dry Raney-Ni cake ignites in air.	F-102 A/B selection; nitrogen through HV-120 for purging	4	3	12	R31 R34

Row	Deviation	Main consequence	Safeguards	S	L	R	Recs
N7.1 †	Temperature MORE: R-103 bed hotter than design (hot spot)	Bed runaway: catalyst sintered, isomer formed, vessel wall above design temperature, loss of containment.	TI-107 HI alarm at 185 °C (hottest of five); TSHH-107 trips at 220 °C (5 s); AI-104 HI alarm at 200 ppm	4	3	12	R37 R01
N7.6	Level LESS: V-104 level lost: gas blow-by to storage	Hydrogen at 29.0 bar g flows through LV-104 into the atmospheric tank TK-106 or TK-107: tank overpressure, rupture, fire.	LIC-104 LO alarm at 25 % (same transmitter as the controller); LV-104 fails closed	5	2	10	R19
N8.2	Flow LESS: Hydrogen flow low with K-101 running	Same as no hydrogen in R-101, but the compressor keeps running so XS-102 does not act: benzene accumulates and passes on to R-102 and R-103; K-101 runs on kickback.	FIC-103 LO alarm at 35000 Nm³/h; K-101 low forward-flow alarm; TIC-101 LO alarm at 190 °C; AI-103 HI alarm at 10 ppm	4	3	12	R24
N8.4 †	Pressure MORE: Loop pressure high	Loop overpressured from the make-up header: loss of containment of hydrogen, fire or explosion.	PIC-105 HI alarm at 30.5 bar g (same transmitter as the controller); PSHH-101 trips at 33.5 bar g (3 s) and closes XV-105	5	2	10	R02 R21
N8.6	Containment OTHER THAN: Hydrogen escapes from the loop	Hydrogen jet fire or explosion near hot equipment; PIC-105 hides the loss by opening PV-105.	PV-105 output and make-up FI-106 rise (hydrogen balance on the Hydrogen loop display); EDV-101 depressuring to flare by the operator	5	2	10	R25
N8.8	Composition AS WELL AS: Oxygen in the loop with hydrogen	Flammable hydrogen–oxygen mixture inside equipment: internal explosion.	AI-100 HI alarm at 1 %; XV-105 permissive AI-100 < 1 %	5	2	10	R26
N9.2	Flow REVERSE: Hydrogen into the nitrogen header	Hydrogen in the plant nitrogen network: fire where the nitrogen is used for inerting.	Nitrogen only flows while the loop is below the header pressure (procedure)	4	3	12	R28
N9.3	Flow NO / NOT: EDV-101 does not open on demand	Loop cannot be depressured in a fire or a leak: vessels weaken under fire exposure while at pressure.	HV-121 vent to flare (manual)	5	2	10	R29
N10.3 †	Time EARLY: Feed ramped up too fast after light-off	Heat release outruns heat removal: TIC-101 overshoots toward the trip.	TIC-101 HI alarm at 208 °C; TSHH-100 trips at 220 °C; XV-101 alarm with R-101 below 180 °C	4	3	12	R32
N10.4 †	Time LATE: HS-101 switched from HEAT to COOL too late	MP steam still heats E-101 while the reaction starts: rapid temperature rise.	TIC-101 HI alarm at 208 °C; TSHH-100 sets HS-101 to COOL and closes HV-110	4	3	12	R16
N10.7	Sequence PART OF: Shutdown purge incomplete before equipment is opened	Flammable release or fire when the flange is broken; pyrophoric catalyst exposed to air.	AI-100 and AI-105 cover the loop only, not the item opened	4	3	12	R34

5.3 Recommendation register

ID	Recommendation	Nodes	Rows	Highest risk	Action by
R01	Add a high-high benzene flow or benzene-to-hydrogen ratio trip (FI-101 against FI-103) that closes XV-101; the feed can run far above design with no trip.	N1 N7	N1.2 N7.1	12 · High	Instrument / Process
R02	Confirm and document the pressure relief of R-101, R-102, R-103 and V-104 for blocked outlet, make-up valve wide open and fire; relief valves are not shown on the training P&ID.	N2 N4 N8	N2.1 N2.7 N4.5 N8.4	12 · High	Process
R03	Add a high-high temperature trip on R-102 with its own element (set below the R-102 boiling point at loop pressure).	N1 N2 N4 N5	N1.2 N2.4 N4.1 N5.1	12 · High	Instrument / Safety
R04	Add low-low agitator power (JI-101) to the SIS so that a running motor with a failed drive is seen like a stopped motor.	N2	N2.4	12 · High	Instrument / Safety
R05	Add an independent R-101 level element with high-high and low-low actions (stop feed; protect P-101A/B).	N2	N2.5 N2.6	9 · Medium	Instrument
R06	Add low cooling-water flow on E-103 or high-high TI-103 to stop the benzene feed.	N4	N4.1 N4.4	12 · High	Process / Instrument
R07	Add independent high-high and low-low level on R-102 (stop P-102A/B, stop feed).	N5	N5.3 N5.4	9 · Medium	Instrument
R08	Fit a non-return valve on the P-100A/B discharge and close XV-101 when no feed pump runs.	N1	N1.4	10 · High	Process / Mechanical
R09	Analyse every benzene receipt for sulphur and water; drain TK-100 water; consider a guard bed ahead of R-101.	N1	N1.7 N1.8	9 · Medium	Process / Operations
R10	Auto-start the standby pumps P-100B, P-101B and P-102B on low discharge pressure or flow.	N1 N3 N6	N1.1 N3.1 N6.1	8 · Medium	Electrical / Instrument

ID	Recommendation	Nodes	Rows	Highest risk	Action by
R11	Add a low-flow alarm on the pump-around FI-104 (none in the alarm list).	N3	N3.1	8 · Medium	Instrument
R12	Add a control-margin alarm on the FIC-104 output and trend the E-101 heat-transfer coefficient.	N3	N3.3	9 · Medium	Process / Instrument
R13	Confirm the D-101 steam-side relief for blocked export at maximum E-101 duty.	N3	N3.4	9 · Medium	Process
R14	Add low-low and high-high level on D-101 independent of LIC-109.	N3	N3.6 N3.7	9 · Medium	Instrument
R15	Monitor D-101 blowdown or condensate for hydrocarbon (E-101 tube leak).	N3	N3.8	8 · Medium	Process
R16	Interlock HV-110 (MP steam to E-101) closed whenever XV-101 is open, and put the HEAT to COOL switch-over in the light-off steps.	N3 N10	N3.9 N10.4	12 · High	Instrument / Operations
R17	Detect hydrogen and hydrocarbon in the CW return of E-103, E-106 and the R-102 jacket (degassing pot or analyser).	N4 N5	N4.6 N5.6	8 · Medium	Process
R18	Route product to TK-107 automatically on AI-103 or AI-104 high and alarm a stale analyser value.	N2 N5 N7 N10	N2.10 N5.2 N7.7 N10.9	6 · Medium	Instrument / Operations
R19	Add an independent low-low level on V-104 closing a shutdown valve in the product line; check the TK-106 / TK-107 vents for gas blow-by.	N7	N7.6	10 · High	Process / Instrument
R20	Add a low ΔP alarm on PDI-102 (torn candle) and a routine fines check of the product.	N6	N6.4	6 · Medium	Instrument / Operations
R21	Add a PI-101 high alarm between the PIC-105 alarm and PSHH-101, because PIC-105 measures at V-104, not at R-101.	N2 N4 N8	N2.7 N4.5 N8.4	10 · High	Instrument
R22	Add a high alarm on TI-109 (V-104 temperature).	N7	N7.3	9 · Medium	Instrument
R23	Add a low-flow alarm on the purge FI-107 and a purity check in the shift routine.	N8	N8.7	9 · Medium	Instrument / Operations
R24	Add low-low hydrogen flow (FI-103) to the SIS to close XV-101; today only a stopped K-101 motor is seen.	N8	N8.2	12 · High	Instrument / Safety
R25	Install hydrogen gas detection at K-101, the reactors and V-104, and alarm the loop hydrogen balance (make-up against 3 × benzene fed).	N8	N8.6	10 · High	Safety / Instrument
R26	Add a second loop oxygen analyser (2oo2) for the XV-105 permissive and record the O2 readings in the purge procedure.	N8 N10	N8.8 N10.1	10 · High	Instrument / Operations
R27	Fit a non-return valve on the K-101 discharge and confirm the kickback / anti-surge protection and its high-temperature alarm.	N8	N8.9 N8.10	9 · Medium	Mechanical
R28	Protect the nitrogen connection: non-return valve, double block and bleed or a removable spool; HV-120 opened only below the header pressure.	N9	N9.2	12 · High	Process / Mechanical
R29	Depressuring study for EDV-101 (rate, flare load, minimum metal temperature); fail-open actuator and proof test.	N9	N9.3 N9.4 N9.7 N9.8	10 · High	Process
R30	Provide a flare knock-out drum and a continuous flare purge or seal to keep air out.	N9	N9.5 N9.6	8 · Medium	Process
R31	Write a Raney-Ni handling procedure: always wet and under nitrogen, metered charges through HV-117, filter-cake handling.	N1 N2 N6 N10	N1.7 N2.9 N6.7 N10.5 N10.6	12 · High	Operations / Safety
R32	Light-off and shutdown steps: limit the feed ramp after XV-101 opens, stop benzene before hydrogen, and add a low-temperature-with-feed interlock (TI-100 with XV-101 open) to stop benzene accumulation.	N2 N3 N10	N2.2 N3.2 N10.3 N10.8	12 · High	Operations / Instrument
R33	Add a TIC-101 against TI-100 deviation alarm (two elements in the same reactor).	N2	N2.1	12 · High	Instrument
R34	Gas-free test and positive isolation before opening any equipment (filters, reactors); written filter change-over procedure.	N6 N10	N6.2 N6.7 N10.7 N10.10	12 · High	Operations
R35	Fit minimum-flow bypasses on P-100A/B and P-102A/B.	N1 N6	N1.1 N1.5 N6.5	9 · Medium	Mechanical
R36	Review the blanketing and venting of TK-106 / TK-107 for hydrogen flashing from the product.	N7	N7.8	9 · Medium	Process
R37	Add an R-103 bed-rise alarm (TI-107E minus TI-106) as early warning of benzene slip.	N1 N7	N1.2 N7.1 N7.9	12 · High	Instrument
R38	Positive segregation of MP steam and CW on the R-102 jacket (interlocked change-over) and jacket relief.	N5	N5.5	6 · Medium	Process / Mechanical

5.4 Dynamic check on the simulator

Ten deviations from the worksheets were forced on copies of the simulator at design and run for 30 minutes with every alarm and trip active and no operator action (results precomputed with the HAZOP tab code; the

tab re-runs them on the page). Times are mm:ss from the start of the deviation.

Run	Node	Deviation (how)	First alarms	Trip (first-out)	Key tags (min → max)
D1	N1	MORE benzene flow (FIC-101 forced to MAN, OUT 100 %)	00:01 FIC-101 HI, 04:12 TIC-101 HI, 04:30 TIC-102 HI, 08:01 AI-103 HI (+4)	none	TI-107 bed max 158.6 → 208.8 °C; TIC-102 160.0 → 206.5 °C; AI-103 2 → 23 728 ppm
D2	N3	NO pump-around flow (FIC-104 forced to MAN, OUT 0 %)	00:57 TIC-101 HI, 01:03 PIC-110 LO, 01:12 TI-100 HI, 05:56 PIC-105 LO	02:57 TSHH-100	TI-100 145.2 → 220.3 °C; FIC-104 0 → 606 t/h; PIC-110 3.24 → 5.18 bar g
D3	N4	NO cooling water (CW supply flow down by 95 % over 30 s)	00:18 TI-103 HI, 01:45 TIC-102 HI, 05:24 TI-107 HI, 16:41 LIC-101 LO	none	TI-103 56.2 → 201.1 °C; TIC-102 160.0 → 196.9 °C; LIC-101 36.6 → 65.0 %
D4	N2	LESS mixing (drive fails, motor runs) (Impeller slows while M-101 stays on)	00:01 JI-101 LO, 00:44 PIC-105 HI, 00:45 TIC-101 LO, 02:00 PIC-110 LO (+5)	none	TI-100 69.6 → 200.0 °C; TIC-102 141.1 → 221.6 °C; AI-103 2 → 18 ppm
D5	N6	NO flow from R-102 to R-103 (LIC-102 forced to MAN, OUT 0 %)	12:35 LIC-102 HI	none	LIC-102 55.0 → 110.0 %; V-104 level 42.1 → 50.0 %; AI-103 1.1 → 2.0 ppm
D6	N7	NO product flow (V-104 fills) (LIC-104 forced to MAN, OUT 0 %)	10:02 LIC-104 HI, 14:25 FIC-103 LO, 19:41 PIC-105 LO, 28:50 PIC-110 LO	14:35 XS-102 (K-101 stopped 14:25)	V-104 level (LSHH element) 50.0 → 95.1 %; PIC-105 22.6 → 30.1 bar g; TI-100 149.5 → 200.0 °C
D7	N8	MORE make-up hydrogen (PIC-105 forced to MAN, OUT 100 %)	00:16 PIC-105 HI, 10:30 PIC-105 LO, 15:04 PIC-110 LO	00:57 PSHH-101	PI-101 18.77 → 33.58 bar g; PIC-105 18.77 → 32.55 bar g
D8	N8	NO hydrogen flow (compressor stops) (K-101 motor trip)	00:01 FIC-103 LO, 05:16 PIC-105 LO, 14:26 PIC-110 LO	00:11 XS-102 (K-101 stopped 00:01)	TI-100 143.2 → 199.8 °C; PIC-105 16.8 → 30.1 bar g; AI-103 2 → 132 ppm
D9	N8	LESS hydrogen flow with K-101 running (FIC-103 forced to MAN, OUT 0 %)	00:01 FIC-103 LO, 00:19 PIC-105 HI, 01:00 TIC-101 LO, 02:20 PIC-110 LO (+1)	none	TI-100 112.2 → 200.0 °C; AI-103 2 → 60 858 ppm; PI-101 28.91 → 31.53 bar g
D10	N2	TIC-101 element reads low (R-101 control element reads 15 °C low over 30 s)	00:21 TIC-101 LO, 02:13 TI-100 HI	none	TI-100 (true) 200.0 → 215.0 °C; AI-104 60 → 175 ppm

What the runs tell the team

- **More benzene has no trip of its own.** With FIC-101 driven to 100 % the R-103 bed reached 208.8 °C (TSHH-107 at 220 °C) and R-102 206.5 °C in 30 min with no trip; the protection was 8 alarms and a product at 23,728 ppm benzene (R01, R03).
- **Two failures the SIS cannot see.** A drive that fails while the motor runs (D4) and a hydrogen valve that closes while K-101 runs (D9) gave no trip in 30 min: R-101 fell to 70 °C and 112 °C with benzene building up, and in D4 R-102 reached 221.6 °C, which has no trip at all. XS-101 and XS-102 watch the motors, not the duty (R03, R04, R24).
- **Loss of pump-around is caught, just.** TSHH-100 tripped at 02:57 after the first alarm (TIC-101 HI at 00:57); TI-100 peaked at 220.3 °C, about two minutes for the operator to act before the SIS does (R11).
- **Losing the reflux condenser does not trip the unit.** TI-103 alarmed at 00:18, R-102 rose to 196.9 °C and R-101 level fell to 36.6 %, below the 40 % start permissive, with no trip in 30 min (R06).
- **A transmitter error moves the reactor close to its trip.** With the TIC-101 element reading low, R-101 settled at 215.0 °C, 5.0 °C under TSHH-100, and only the independent TI-100 alarm (02:13) showed it; isomer rose to 175 ppm against the 200 ppm spec (R33).
- **The existing trips work where they look.** Make-up wide open: PSHH-101 at 00:57 with PI-101 peaking at 33.58 bar g (set 33.5); K-101 motor stop: XS-102 at 00:11; V-104 filling: LSHH-104 stopped K-101 at 14:25 and XS-102 tripped at 14:35. A trip only limits the deviation it measures; relief is still needed for faster events (R02).
- **R-102 can fill with one alarm.** With LV-102 shut, R-102 reached 110 % (top of range) after a single LIC-102 HI alarm at 12:35 and no trip (R07).

5.5 Follow-up

LOPA for the High rows that rely on a trip (the trip becomes a candidate SIF; its SIL comes from LOPA and is recorded in the SRS CHX-V1-IN-SRS-001):

- **N1.2** Flow MORE: More benzene than the hydrogen and heat removal can take (TSHH-100, TSHH-107)
- **N2.1** Temperature MORE: R-101 above 200.0 °C toward its boiling point (about 258 °C at reactor pressure) (TSHH-100, PSHH-101)
- **N2.7** Pressure MORE: R-101 pressure above normal (PSHH-101)

- **N4.1** Flow NO / NOT: No cooling water to E-103 (TSHH-107)
- **N5.1** Temperature MORE: R-102 hotter than 160.0 °C (TSHH-107)
- **N7.1** Temperature MORE: R-103 bed hotter than design (hot spot) (TSHH-107)
- **N8.4** Pressure MORE: Loop pressure high (PSHH-101)
- **N10.3** Time EARLY: Feed ramped up too fast after light-off (TSHH-100)
- **N10.4** Time LATE: HS-101 switched from HEAT to COOL too late (TSHH-100)

Responses and close-out: each action party answers its recommendations on the close-out sheets of Appendix 3 (accept, accept with modification, or reject with reasons); the HAZOP leader accepts the response, and the register stays open until every action is closed. Target dates by risk (typical practice): High before the next start-up, Medium at the next turnaround, Low when convenient.

ID	Action by	Highest risk	Target	Status
R01	Instrument / Process	12 · High	Before the next start-up	OPEN
R02	Process	12 · High	Before the next start-up	OPEN
R03	Instrument / Safety	12 · High	Before the next start-up	OPEN
R04	Instrument / Safety	12 · High	Before the next start-up	OPEN
R05	Instrument	9 · Medium	Next turnaround	OPEN
R06	Process / Instrument	12 · High	Before the next start-up	OPEN
R07	Instrument	9 · Medium	Next turnaround	OPEN
R08	Process / Mechanical	10 · High	Before the next start-up	OPEN
R09	Process / Operations	9 · Medium	Next turnaround	OPEN
R10	Electrical / Instrument	8 · Medium	Next turnaround	OPEN
R11	Instrument	8 · Medium	Next turnaround	OPEN
R12	Process / Instrument	9 · Medium	Next turnaround	OPEN
R13	Process	9 · Medium	Next turnaround	OPEN
R14	Instrument	9 · Medium	Next turnaround	OPEN
R15	Process	8 · Medium	Next turnaround	OPEN
R16	Instrument / Operations	12 · High	Before the next start-up	OPEN
R17	Process	8 · Medium	Next turnaround	OPEN
R18	Instrument / Operations	6 · Medium	Next turnaround	OPEN
R19	Process / Instrument	10 · High	Before the next start-up	OPEN
R20	Instrument / Operations	6 · Medium	Next turnaround	OPEN
R21	Instrument	10 · High	Before the next start-up	OPEN
R22	Instrument	9 · Medium	Next turnaround	OPEN
R23	Instrument / Operations	9 · Medium	Next turnaround	OPEN
R24	Instrument / Safety	12 · High	Before the next start-up	OPEN
R25	Safety / Instrument	10 · High	Before the next start-up	OPEN
R26	Instrument / Operations	10 · High	Before the next start-up	OPEN
R27	Mechanical	9 · Medium	Next turnaround	OPEN
R28	Process / Mechanical	12 · High	Before the next start-up	OPEN
R29	Process	10 · High	Before the next start-up	OPEN
R30	Process	8 · Medium	Next turnaround	OPEN
R31	Operations / Safety	12 · High	Before the next start-up	OPEN
R32	Operations / Instrument	12 · High	Before the next start-up	OPEN
R33	Instrument	12 · High	Before the next start-up	OPEN
R34	Operations	12 · High	Before the next start-up	OPEN
R35	Mechanical	9 · Medium	Next turnaround	OPEN
R36	Process	9 · Medium	Next turnaround	OPEN
R37	Instrument	12 · High	Before the next start-up	OPEN
R38	Process / Mechanical	6 · Medium	Next turnaround	OPEN

Appendix 1 Worksheets

A1.1 Node N1 · Benzene feed

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N1 · Benzene feed	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Deliver 34.8 t/h of benzene at 40 °C from TK-100 by P-100A/B under FIC-101 control through FV-101 and the SIS valve XV-101 into R-101 at 30.0 bar g.		
Design conditions	Benzene feed 34.8 t/h · Feed temperature 40 °C · R-101 pressure (delivery) 30.0 bar g (simulator design state)		
Equipment	TK-100, P-100A/B (one running, one standby), FV-101, XV-101	Instruments	FIC-101 (FV-101 fails closed), XV-101 (SIS)

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO / NOT	No benzene flow to R-101	• Running feed pump P-100A stops (motor trip) and P-100B is not started • FV-101 closes (fails closed on loss of instrument air) • XV-101 closed by the SIS or spuriously • TK-100 empty or suction strainer blocked	Reaction stops: R-101 cools, hydrogen consumption falls and the loop pressure rises until PIC-105 closes PV-105; production lost and a new light-off needed. A pump running against a closed FV-101 or XV-101 overheats and its seal can leak benzene.	• FIC-101 LO alarm at 25 t/h • standby pump P-100B • TIC-101 LO alarm at 190 °C • PIC-105 holds loop pressure by closing PV-105	2	3	**6 Medium**	R10 R35	Electrical / Instrument · Mechanical
N1.2	Flow	MORE	More benzene than the hydrogen and heat removal can take	• FIC-101 output driven high (controller or positioner fault, set-point entry error) • FI-101 transmitter reading low • FV-101 trim damaged	Conversion in R-101 falls and unreacted benzene passes to R-102 and R-103, which then release its heat of reaction: R-102 and the adiabatic R-103 bed heat up (about 9 °C per 1 % benzene slip), the product goes off-spec and the bed can run away.	• FIC-101 HI alarm at 42 t/h • TIC-102 HI alarm at 172 °C • TI-107 HI alarm at 185 °C • TSHH-107 trips at 220 °C (5 s) • TSHH-100 trips at 220 °C (5 s) • AI-103 HI alarm at 10 ppm	4	3	**12 High**	R01 R03 R37	Instrument / Process · Instrument / Safety · Instrument
N1.3	Flow	LESS	Less benzene than set	• FV-101 partly stuck • P-100A/B worn or cavitating • suction strainer partly blocked	Lower rate and margin; R-101 runs with more hydrogen excess. No hazard.	• FIC-101 LO alarm at 25 t/h • FIC-101 in AUTO opens FV-101 further	1	4	**4 Low**	—	—
N1.4	Flow	REVERSE	Reactor liquid and hydrogen flow back toward P-100A/B and TK-100	• Feed pump stops while R-101 is at 30.0 bar g with FV-101 and XV-101 open • no non-return valve shown on the pump discharge	Hot liquid saturated with hydrogen enters the feed line and the atmospheric tank TK-100: tank overpressure, release of hydrogen and benzene vapour, fire.	• XV-101 closes on any SIS trip • FV-101 fails closed • FIC-101 LO alarm at 25 t/h	5	2	**10 High**	R08	Process / Mechanical
N1.5	Pressure	MORE	Feed line at pump shut-off pressure (dead-headed pump)	• XV-101 closed by the SIS or FV-101 closed while P-100A runs • a manual valve downstream left shut	Pump runs at shut-off: liquid heats, seal and bearing damage, benzene leak at the seal (flammable, carcinogenic).	• FIC-101 LO alarm at 25 t/h • trip procedure stops the feed pump	3	3	**9 Medium**	R35	Mechanical
N1.6	Temperature	MORE	Benzene feed hotter than the design 40 °C	• TK-100 heated by sun or tracing left on • product recycled to the tank hot	Slightly less heat absorbed by the feed (TIC-101 compensates); pump NPSH margin falls, cavitation.	• TIC-101 cascade to FIC-104 compensates • TIC-101 HI alarm at 208 °C	1	2	**2 Low**	—	—
N1.7	Composition	OTHER THAN	Sulphur compounds (thiophene) in the benzene	• Off-spec benzene cargo • TK-100 contaminated • wrong tank lined up	The Raney-Ni in R-101 is poisoned irreversibly: conversion falls, benzene slips to R-102 and R-103 (more heat there), product off-spec; the catalyst must be replaced.	• AI-103 HI alarm at 10 ppm • TI-107 HI alarm at 185 °C • fresh catalyst through the catalyst pot HV-117	3	3	**9 Medium**	R09 R31	Process / Operations · Operations / Safety
N1.8	Composition	AS WELL AS	Water with the benzene	• Tank bottoms drawn into the suction • water not drained from TK-100	Water flashes in R-101 at 200 °C: pressure surge, catalyst activity lost, water carried to V-104 and the product.	• TIC-101 and PIC-105 control • PSHH-101 trips at 33.5 bar g (3 s)	2	3	**6 Medium**	R09	Process / Operations

A1.2 Node N2 · R-101 slurry reactor

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N2 · R-101 slurry reactor	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Hydrogenate about 98 % of the benzene in a stirred Raney-Ni slurry at 200.0 °C and 30.0 bar g with 65.0 % level, agitator M-101 dispersing the hydrogen from the sparger; liquid overflows by LIC-101 / LV-101 to R-102.		
Design conditions	Temperature TIC-101 200.0 °C · Pressure PI-101 30.0 bar g · Level LIC-101 65.0 % · Agitator power JI-101 153 kW · Heat of reaction released 25.0 MW · Boiling point at reactor pressure 258 °C (simulator design state)		
Equipment	R-101, agitator M-101, hydrogen sparger, catalyst pot HV-117, LV-101	Instruments	TIC-101 (cascade to FIC-104), TI-100 (independent element), PI-101, LIC-101 (LV-101 fails closed), JI-101; TSHH-100, PSHH-101, XS-101

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Temperature	MORE	R-101 above 200.0 °C toward its boiling point (about 258 °C at reactor pressure)	• E-101 heat removal lost or reduced (see N3) • reflux condenser lost (see N4) • TIC-101 measurement reading low • TIC-101 left in MAN • too much catalyst charged • feed rate too high	Faster reaction and more isomer (MCP); at the boiling ceiling the extra heat becomes vapour, the pressure and the condenser load rise and hot hydrocarbon is carried to R-102. Overpressure of R-101 if the vapour cannot leave.	• TIC-101 HI alarm at 208 °C • TI-100 independent element HI alarm at 210 °C • TSHH-100 trips at 220 °C (5 s): XV-101 and XV-105 shut, K-101 stops, E-101 to COOL with FV-104 fully open • PSHH-101 trips at 33.5 bar g	4	3	**12 High**	R02 R33	Process · Instrument
N2.2	Temperature	LESS	R-101 below the light-off temperature with feed in	• Over-cooling (FV-104 fully open, which is its fail position) • catalyst poisoned or aged • hydrogen flow low • agitation reduced	The reaction slows while benzene keeps coming: unreacted benzene accumulates in R-101 and slips to R-102. When the cause is removed the accumulated benzene reacts at once and the temperature can jump past the trip.	• TIC-101 LO alarm at 190 °C • XV-101 alarm when benzene is permitted with R-101 below 180 °C • feed permissive TI-100 ≥ 140 °C (start only)	4	3	**12 High**	R32	Operations / Instrument
N2.3	Mixing	NO / NOT	Agitator M-101 stops	• M-101 motor trip • electrical supply interruption • stop button pressed	Catalyst settles and gas no longer disperses: reaction in R-101 stops, benzene accumulates and slips to R-102.	• XS-101 trips the unit when M-101 stops with XV-101 open (10 s) • JI-101 LO alarm at 80 kW	4	2	**8 Medium**	—	—
N2.4	Mixing	LESS	Impeller turns slowly while the motor runs	• Coupling or shaft failure • impeller loose or lost	As for no agitation, but the motor-run signal stays healthy so XS-101 does not act: benzene builds up in R-101 and R-102 receives it; R-102 has no high-high temperature trip.	• JI-101 LO alarm at 80 kW • TIC-101 LO alarm at 190 °C • TIC-102 HI alarm at 172 °C	4	3	**12 High**	R04 R03	Instrument / Safety
N2.5	Flow	NO / NOT	No overflow from R-101 to R-102	• LV-101 stuck or failed closed (loss of instrument air) • LIC-101 transmitter reads low • LIC-101 left in MAN	R-101 level rises: less vapour space, liquid and foam carried into the vapour line to E-103 / V-103 and with the gas to R-102; downstream reactors starve.	• LIC-101 HI alarm at 82 % • LIC-101 in AUTO	3	3	**9 Medium**	R05	Instrument
N2.6	Level	LESS	R-101 level low	• LIC-101 output high or transmitter reads high • boil-off not returned (reflux lost, see N4) • feed lost with LV-101 open	Sparger and agitator uncovered (start permissive is 40 %), pump-around P-101A/B loses suction and cavitates; less liquid to absorb heat.	• LIC-101 LO alarm at 45 % • level permissive ≥ 40 % before feed (start only)	3	3	**9 Medium**	R05	Instrument
N2.7	Pressure	MORE	R-101 pressure above normal	• Make-up valve PV-105 driven open (PIC-105 output or transmitter fault) • restriction in the series gas path R-101 → E-103 / V-103 → R-102 → R-103 → V-104 • boiling at the ceiling	Overpressure of R-101 and the loop: loss of containment of hydrogen and hot hydrocarbon, fire or explosion.	• PIC-105 HI alarm at 30.5 bar g (measured at V-104) • PSHH-101 trips at 33.5 bar g (3 s): XV-105 make-up and XV-101 shut, K-101 stops	5	2	**10 High**	R02 R21	Process · Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N2.8	Pressure	LESS	R-101 pressure below normal	• Loop pressure lost (make-up lost, leak, EDV-101 open) • PIC-105 transmitter reading high	Liquid at 200 °C flashes: level swells, carry-over to E-103 and R-102, lower hydrogen partial pressure and conversion.	• PIC-105 LO alarm at 27.5 bar g • AI-103 HI alarm at 10 ppm	2	3	**6 Medium**	—	—
N2.9	Reaction	MORE	More catalyst activity than intended	• Catalyst pot HV-117 left open (catalyst added continuously) • fresh catalyst more active than the aged charge	Reaction rises faster than TIC-101 can follow: temperature overshoot, more isomer.	• TIC-101 cascade • TIC-101 HI alarm at 208 °C • TSHH-100 trips at 220 °C	3	2	**6 Medium**	R31	Operations / Safety
N2.10	Composition	AS WELL AS	Isomer (MCP) formed with the cyclohexane	• R-101 running hot • R-103 bed hot	Product off-spec on MCP (spec 200 ppm) sent to the on-spec tank.	• AI-104 HI alarm at 200 ppm • HS-115 routes product to TK-107	1	4	**4 Low**	R18	Instrument / Operations

A1.3 Node N3 · Pump-around, E-101 and steam drum

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N3 · Pump-around, E-101 and steam drum	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Circulate 357.3 t/h of reactor slurry by P-101A/B through E-101 (FIC-104 in cascade from TIC-101, FV-104 fails open) and remove 9.6 MW by raising LP steam in D-101 at 4.50 bar g; HS-101 HEAT puts MP steam (HV-110) on E-101 for start-up only.		
Design conditions	Pump-around FIC-104 357 t/h · Return temperature TI-104 159.3 °C · E-101 duty 9.6 MW · D-101 pressure PIC-110 4.50 bar g · D-101 level LIC-109 50 % · LP steam export FI-111 14.7 t/h (simulator design state)		
Equipment	P-101A/B, E-101, FV-104, D-101, LV-109, PV-110, HV-110, HS-101	Instruments	FIC-104 (FV-104 fails open), TI-104, PIC-110 (PV-110 fails open), LIC-109 (LV-109 fails open), FI-111, FI-109

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	NO / NOT	No pump-around flow through E-101	• P-101A stops (motor trip) and P-101B is not started • R-101 level below the pump suction • FIC-104 output driven to 0 % (controller fault)	E-101 removes no heat (design 9.6 MW): R-101 heats toward the trip and the boiling ceiling; D-101 steam make falls and LP steam export stops.	• TIC-101 HI alarm at 208 °C • TI-100 HI alarm at 210 °C • TSHH-100 trips at 220 °C (5 s) • FV-104 fails open • standby pump P-101B	4	2	**8 Medium**	R10 R11	Electrical / Instrument · Instrument
N3.2	Flow	MORE	Pump-around flow at maximum	• FV-104 fails open (loss of instrument air) or FIC-104 output high • TIC-101 measurement reads high	R-101 over-cooled below light-off: benzene accumulates (see N2); D-101 makes more steam and its pressure rises.	• TIC-101 LO alarm at 190 °C • XV-101 alarm with R-101 below 180 °C • PIC-110 HI alarm at 5.6 bar g	3	3	**9 Medium**	R32	Operations / Instrument
N3.3	Temperature	MORE	Pump-around return TI-104 hotter than design	• E-101 fouled by polymer and catalyst fines • low water level in D-101 uncovering tubes	Heat-transfer margin used up slowly while TIC-101 hides it by opening FV-104; once FV-104 is wide open R-101 temperature rises and the unit trips or must cut rate.	• TIC-101 HI alarm at 208 °C • FIC-104 output shown on the faceplate • TSHH-100 trips at 220 °C	3	3	**9 Medium**	R12	Process / Instrument
N3.4	Pressure	MORE	D-101 pressure above 4.50 bar g	• High LP steam header pressure (a large user trips) so PV-110 cannot export • PV-110 stuck closed • PIC-110 transmitter reads low	Higher saturation temperature in D-101 lowers E-101 duty: R-101 heats; drum overpressure if export stays blocked.	• PIC-110 HI alarm at 5.6 bar g • PV-110 fails open • TIC-101 HI alarm at 208 °C • TSHH-100 trips at 220 °C	3	3	**9 Medium**	R13	Process
N3.5	Pressure	LESS	D-101 pressure below design	• PV-110 fails open (loss of instrument air) or PIC-110 output high	Lower saturation temperature raises E-101 duty: R-101 is cooled harder (TIC-101 compensates); wet steam to the header.	• PIC-110 LO alarm at 3.8 bar g • TIC-101 cascade	2	3	**6 Medium**	—	—
N3.6	Level	LESS	D-101 water level low	• Boiler feed water lost • LV-109 closed in MAN • LIC-109 transmitter reads high	E-101 tubes uncovered: heat removal falls (R-101 heats) and refilling a dry hot drum causes thermal shock.	• LIC-109 LO alarm at 25 % • LV-109 fails open	3	3	**9 Medium**	R14	Instrument
N3.7	Level	MORE	D-101 water level high	• LV-109 fails open (loss of instrument air) • LIC-109 transmitter reads low	Water carried into the LP steam header: water hammer in steam users.	• LIC-109 HI alarm at 75 %	3	3	**9 Medium**	R14	Instrument
N3.8	Composition	AS WELL AS	Hydrocarbon and hydrogen in the steam system	• E-101 tube leak: the process side (30.0 bar g) is above the steam side (4.50 bar g)	Cyclohexane, benzene and hydrogen enter D-101 and the LP steam header: flammable release at steam users and vents.	• None on the P&ID	4	2	**8 Medium**	R15	Process
N3.9	Utility	OTHER THAN	MP steam on E-101 while reacting (HS-101 on HEAT)	• HS-101 not switched from HEAT to COOL after light-off • HV-110 left open	E-101 heats R-101 instead of cooling it: temperature rises quickly at full rate.	• TIC-101 HI alarm at 208 °C • TSHH-100 trips at 220 °C and sets HS-101 to COOL with HV-110 closed	4	2	**8 Medium**	R16	Instrument / Operations

A1.4 Node N4 · Reflux condenser E-103 / V-103

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N4 · Reflux condenser E-103 / V-103	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Condense the cyclohexane boiled off R-101 with the hydrogen in E-103 on cooling water (HV-113), return 46.5 t/h of reflux from V-103 to R-101 and pass the gas at 55.7 °C to R-102.		
Design conditions	Reflux FI-105 46.5 t/h · Vent temperature TI-103 55.7 °C · CW supply TI-120 32 °C · Boil-up condensed 3.3 MW (simulator design state)		
Equipment	E-103, V-103, HV-113, vent line V-103 → R-102	Instruments	TI-103, FI-105, TI-120

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Flow	NO / NOT	No cooling water to E-103	• CW supply lost (pump trip at the cooling tower) • HV-113 closed in error	Reflux FI-105 stops: R-101 loses a heat sink and its level falls; hot cyclohexane vapour goes with the hydrogen to R-102, which heats, and on to R-103.	• TI-103 HI alarm at 75 °C • LIC-101 LO alarm at 45 % • TIC-102 HI alarm at 172 °C • TSHH-107 trips at 220 °C • feed permissive HV-113 open (start only)	4	3	**12 High**	R06 R03	Process / Instrument · Instrument / Safety
N4.2	Temperature	MORE	CW supply hotter than 32.0 °C	• Cooling tower performance falls in hot weather or with fans out	E-103 vent runs warmer: more cyclohexane vapour leaves with the hydrogen to R-102 and less reflux returns.	• TI-120 indication • TI-103 HI alarm at 75 °C	2	4	**8 Medium**	—	—
N4.3	Flow	LESS	Reflux return from V-103 reduced	• V-103 return line partly blocked by polymer or catalyst fines	V-103 backs up and liquid is carried with the vent gas to R-102; R-101 level falls slowly.	• FI-105 indication • LIC-101 LO alarm at 45 % • LIC-102 HI alarm at 80 %	2	2	**4 Low**	—	—
N4.4	Flow	MORE	Vapour load above condenser capacity	• R-101 at its boiling ceiling (heat removal lost elsewhere) • hydrogen flow raised	Uncondensed vapour passes to R-102: R-102 and R-103 heat.	• TI-103 HI alarm at 75 °C • TIC-101 HI alarm at 208 °C • TSHH-100 trips at 220 °C	3	3	**9 Medium**	R06	Process / Instrument
N4.5	Pressure	MORE	R-101 pressure rises against a restricted vent path	• Vent line V-103 → R-102 fouled or a valve in it partly closed	R-101 pressure rises while PIC-105, which measures at V-104, sees nothing: R-101 relies on PSHH-101 alone.	• PSHH-101 trips at 33.5 bar g (3 s) • PI-101 indication (no alarm)	4	2	**8 Medium**	R21 R02	Instrument · Process
N4.6	Composition	AS WELL AS	Hydrogen and hydrocarbon in the cooling water	• E-103 tube leak: process side (30.0 bar g) above the CW side	Hydrogen and cyclohexane carried to the cooling tower: flammable cloud at the tower, fire.	• None on the P&ID	4	2	**8 Medium**	R17	Process
N4.7	Flow	REVERSE	Liquid from R-102 back into V-103 / E-103	• R-101 pressure falls below R-102 (R-101 depressured first, gas flow reversed after K-101 stops)	R-102 slurry backs into the condenser and reflux line: catalyst fouling of E-103.	• Gas flows in series from K-101, which keeps R-101 at the highest pressure in normal operation	2	2	**4 Low**	—	—

A1.5 Node N5 · R-102 and its jacket

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N5 · R-102 and its jacket	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Hydrogenate most of the remaining benzene at 160.0 °C and 29.6 bar g, with the jacket on MP steam (HS-102 HEAT, TIC-102 → TV-102) or cooling water (COOL); hold 55.0 % level.		
Design conditions	Temperature TIC-102 160.0 °C · Pressure PI-102 29.6 bar g · Level LIC-102 55.0 % · Jacket MP steam FI-108 2.33 t/h (simulator design state)		
Equipment	R-102, jacket, TV-102, HS-102	Instruments	TIC-102 (TV-102 fails closed), LIC-102, PI-102, FI-108

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Temperature	MORE	R-102 hotter than 160.0 °C	• Benzene slip from R-101 (more feed, low activity or poor mixing there) • hot vapour from E-103 when its cooling is lost • HS-102 left on HEAT when cooling is needed • TV-102 output high in HEAT	More vapour and benzene go on to R-103, whose adiabatic bed heats; R-102 approaches its boiling ceiling and has no high-high trip of its own.	• TIC-102 HI alarm at 172 °C • TV-102 fails closed (no steam in HEAT) • any SIS trip sets HS-102 to COOL • TSHH-107 trips at 220 °C	4	3	**12 High**	R03	Instrument / Safety
N5.2	Flow	NO / NOT	No heating medium in the R-102 jacket	• MP steam lost • TV-102 fails closed (loss of instrument air) • HS-102 on COOL in error	R-102 cools: conversion there falls and more benzene reaches R-103; product can go off-spec.	• TIC-102 LO alarm at 148 °C • R-103 finishes the benzene • AI-103 HI alarm at 10 ppm	2	3	**6 Medium**	R18	Instrument / Operations
N5.3	Level	MORE	R-102 level high	• LV-102 closed or failed closed • P-102A stops • F-102 blinded • LIC-102 transmitter reads low	Vapour space lost and liquid carried with the gas to R-103 (slug on the bed); R-101 overflow keeps filling R-102.	• LIC-102 HI alarm at 80 % • PDI-102 HI alarm at 0.8 bar	3	3	**9 Medium**	R07	Instrument
N5.4	Level	LESS	R-102 level low	• LIC-102 transmitter reads high • LV-102 output high	P-102A/B run dry or cavitate; gas blows through the filter into the R-103 liquid inlet.	• LIC-102 LO alarm at 30 %	2	3	**6 Medium**	R07	Instrument
N5.5	Pressure	MORE	Jacket at MP steam pressure on the CW side	• HS-102 change-over leaves steam and CW connected together • TV-102 lets steam into the CW return	CW return and jacket overpressured, steam hammer in the CW header.	• HS-102 selects one medium at a time	3	2	**6 Medium**	R38	Process / Mechanical
N5.6	Composition	AS WELL AS	Hydrocarbon in the jacket medium	• Jacket wall leak: process (29.6 bar g) above steam (10 bar g) or CW	Hydrogen and hydrocarbon into condensate or cooling water: flammable release at the tower or the condensate tank.	• None on the P&ID	3	2	**6 Medium**	R17	Process
N5.7	Composition	PART OF	Hydrogen to R-102 short	• FIC-103 low • R-101 consumes almost all the hydrogen • purity low (see N8)	R-102 starves: benzene leaving R-102 rises and R-103 must finish it; product off-spec.	• FIC-103 LO alarm at 35000 Nm³/h • AI-105 LO alarm at 85 % • AI-103 HI alarm at 10 ppm	2	3	**6 Medium**	—	—

A1.6 Node N6 · Transfer and catalyst filter

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N6 · Transfer and catalyst filter	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Pump 20.9 t/h of R-102 liquid by P-102A/B through the candle filter F-102A or B (PDI-102 0.26 bar clean) and LV-102 to R-103, keeping the Raney-Ni in R-102.		
Design conditions	Liquid to R-103 FI-102 20.9 t/h · Filter ΔP PDI-102 0.26 bar (simulator design state)		
Equipment	P-102A/B, F-102A/B, LV-102	Instruments	PDI-102, FI-102, LIC-102 (LV-102 fails closed)

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	NO / NOT	No liquid from R-102 to R-103	• P-102A stops (motor trip) and P-102B is not started • LV-102 fails closed (loss of instrument air) • F-102 fully blinded	R-102 fills (see N5); R-103 receives vapour only and the product rate to V-104 falls.	• LIC-102 HI alarm at 80 % • LIC-104 LO alarm at 25 % • standby pump P-102B	2	3	**6 Medium**	R10	Electrical / Instrument
N6.2	Flow	LESS	Filter F-102 restricts the flow	• Catalyst fines blind the candles of the filter in service	PDI-102 rises, LV-102 opens to compensate and then R-102 level rises.	• PDI-102 HI alarm at 0.8 bar • standby filter selected by F-102 A/B	2	4	**8 Medium**	R34	Operations
N6.3	Flow	MORE	Surge of liquid to R-103	• LIC-102 transmitter reads low • LV-102 output high	R-102 drained, gas blows through to the R-103 inlet; bed temperature profile shifts.	• LIC-102 LO alarm at 30 %	2	2	**4 Low**	—	—
N6.4	Composition	AS WELL AS	Catalyst fines with the liquid to R-103	• Torn filter candle or gasket bypass on F-102	Raney-Ni fines foul the R-103 bed and reach V-104 and the product tanks (pyrophoric solids in storage).	• PDI-102 indication (a torn candle shows a LOW ΔP, which has no alarm)	3	2	**6 Medium**	R20	Instrument / Operations
N6.5	Pressure	MORE	Pump discharge at shut-off pressure	• LV-102 closed with P-102A running	Pump overheating and seal leak of hot cyclohexane with hydrogen.	• LIC-102 HI alarm at 80 %	2	3	**6 Medium**	R35	Mechanical
N6.6	Temperature	MORE	Hot slurry through the filter	• R-102 running hot (see N5)	Filter gaskets and candles above their rating: fines bypass.	• TIC-102 HI alarm at 172 °C	2	2	**4 Low**	—	—
N6.7	Maintenance	OTHER THAN	Filter opened for cleaning while not isolated or not gas-free	• Change-over done with leaking isolation • hydrogen and hydrocarbon not purged • filter cake allowed to dry	Hydrogen and hot hydrocarbon released when the filter is opened; dry Raney-Ni cake ignites in air.	• F-102 A/B selection • nitrogen through HV-120 for purging	4	3	**12 High**	R31 R34	Operations / Safety · Operations

A1.7 Node N7 · R-103, E-106, V-104 and product

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N7 · R-103, E-106, V-104 and product	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Finish the benzene to about 2.00 ppm in the adiabatic trickle bed R-103 (158.6 °C outlet), cool in E-106, separate gas and liquid in V-104 at 29.0 bar g and 44.9 °C, and send 37.5 t/h of product by LIC-104 / LV-104 to TK-106 (on spec) or TK-107 (slop) by HS-115.		
Design conditions	Bed outlet TI-107E 158.6 °C · Product benzene AI-103 2.0 ppm · V-104 pressure PIC-105 29.0 bar g · V-104 temperature TI-109 44.9 °C · V-104 level LIC-104 50 % · Product FI-110 37.5 t/h (simulator design state)		
Equipment	R-103, E-106, V-104, LV-104, HS-115, TK-106, TK-107	Instruments	TI-106, TI-107A to TI-107E, TI-109, LIC-104 (LV-104 fails closed), LSHH-104, AI-103, AI-104, FI-110; TSHH-107

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Temperature	MORE	R-103 bed hotter than design (hot spot)	• Benzene slip from R-102 (each 1 % benzene gives about 9 °C adiabatic rise) • hot inlet from R-102	Bed runaway: catalyst sintered, isomer formed, vessel wall above design temperature, loss of containment.	• TI-107 HI alarm at 185 °C (hottest of five) • TSHH-107 trips at 220 °C (5 s) • AI-104 HI alarm at 200 ppm	4	3	**12 High**	R37 R01	Instrument · Instrument / Process
N7.2	Temperature	LESS	R-103 inlet colder than design	• R-102 cooled (jacket heating lost) • low flow	Finishing activity falls: benzene in product rises.	• TI-106 indication • AI-103 HI alarm at 10 ppm	1	3	**3 Low**	—	—
N7.3	Temperature	MORE	V-104 separator hot	• E-106 cooling water lost or reduced • warm CW	More cyclohexane vapour to K-101 and the purge; hotter compressor suction; product flashes in storage.	• TI-109 indication (no alarm)	3	3	**9 Medium**	R22	Instrument
N7.4	Flow	NO / NOT	No product flow to storage	• LV-104 fails closed (loss of instrument air) • tank inlet valve shut • HS-115 lined to a full tank	V-104 fills toward the K-101 suction (next row).	• LIC-104 HI alarm at 78 % • LSHH-104 stops K-101 at 90 % (5 s)	3	3	**9 Medium**	—	—
N7.5	Level	MORE	V-104 level high, liquid toward K-101	• LIC-104 transmitter reading low (the controller then closes LV-104) • LIC-104 left in MAN	Liquid carried into K-101: compressor damage, release of hydrogen.	• LSHH-104 stops K-101 at 90 % on its own level element (5 s), then XS-102 trips the unit (10 s) • LIC-104 HI alarm at 78 % (same transmitter as the controller)	4	2	**8 Medium**	—	—
N7.6	Level	LESS	V-104 level lost: gas blow-by to storage	• LIC-104 transmitter reads high • LV-104 output high • LIC-104 in MAN with LV-104 open	Hydrogen at 29.0 bar g flows through LV-104 into the atmospheric tank TK-106 or TK-107: tank overpressure, rupture, fire.	• LIC-104 LO alarm at 25 % (same transmitter as the controller) • LV-104 fails closed	5	2	**10 High**	R19	Process / Instrument
N7.7	Composition	OTHER THAN	Off-spec product sent to the on-spec tank	• Benzene slip • analyser AI-103 frozen on an old value • HS-115 not switched to TK-107	TK-106 contaminated with benzene above 10 ppm: tank downgraded, customer claim.	• AI-103 HI alarm at 10 ppm • AI-104 HI alarm at 200 ppm • HS-115 to TK-107	2	3	**6 Medium**	R18	Instrument / Operations
N7.8	Pressure	MORE	Storage tank pressure raised by dissolved hydrogen	• Hydrogen dissolved at 29.0 bar g flashes when the product reaches the atmospheric tank	Hydrogen in the tank vapour space: flammable atmosphere, vent load.	• None on the P&ID	3	3	**9 Medium**	R36	Process
N7.9	Reaction	LESS	R-103 finishing activity low	• Finishing catalyst deactivated or poisoned	Benzene in product rises; the bed temperature rise disappears.	• AI-103 HI alarm at 10 ppm • TI-107A to TI-107E profile	2	3	**6 Medium**	R37	Instrument

A1.8 Node N8 · Hydrogen loop

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N8 · Hydrogen loop	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Recycle 46,000 Nm³/h of hydrogen by K-101 (discharge 32.0 bar g) through FIC-103 / FV-103 to the R-101 sparger; make up 31,974 Nm³/h by PIC-105 / PV-105 holding V-104 at 29.0 bar g; purge 2,017 Nm³/h by HIC-107 to hold purity at 92.1 %.		
Design conditions	Recycle FIC-103 46 000 Nm³/h · K-101 discharge PI-112 32.0 bar g · Loop pressure PIC-105 29.0 bar g · Make-up FI-106 31 974 Nm³/h · Purge FI-107 2 017 Nm³/h · Purity AI-105 92.1 % (simulator design state)		
Equipment	K-101, FV-103, PV-105, XV-105, HIC-107, loop piping R-101 → R-102 → R-103 → V-104 → K-101	Instruments	FIC-103 (FV-103 fails closed), PIC-105 (PV-105 fails closed), PI-112, AI-105, AI-100, FI-106, FI-107; XS-102, PSHH-101

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Flow	NO / NOT	No hydrogen flow (compressor stopped)	• K-101 motor trip • loss of electrical supply • LSHH-104 action	All three reactors lose hydrogen: reactions stop and benzene accumulates if the feed continues; loop pressure decays through the purge.	• XS-102 trips the unit when K-101 stops with XV-101 open (10 s) • FIC-103 LO alarm at 35000 Nm³/h	4	2	**8 Medium**	—	—
N8.2	Flow	LESS	Hydrogen flow low with K-101 running	• FV-103 fails closed (loss of instrument air) or FIC-103 output low • FI-103 transmitter reads high	Same as no hydrogen in R-101, but the compressor keeps running so XS-102 does not act: benzene accumulates and passes on to R-102 and R-103; K-101 runs on kickback.	• FIC-103 LO alarm at 35000 Nm³/h • K-101 low forward-flow alarm • TIC-101 LO alarm at 190 °C • AI-103 HI alarm at 10 ppm	4	3	**12 High**	R24	Instrument / Safety
N8.3	Flow	MORE	Hydrogen flow high	• FIC-103 output high • FI-103 transmitter reads low	More gas strips more cyclohexane from R-101: higher condenser load and more vapour to R-102.	• FIC-103 HI alarm at 56000 Nm³/h • TI-103 HI alarm at 75 °C	2	3	**6 Medium**	—	—
N8.4	Pressure	MORE	Loop pressure high	• PV-105 driven open (PIC-105 output high or its transmitter reads low) • make-up supply pressure high	Loop overpressured from the make-up header: loss of containment of hydrogen, fire or explosion.	• PIC-105 HI alarm at 30.5 bar g (same transmitter as the controller) • PSHH-101 trips at 33.5 bar g (3 s) and closes XV-105	5	2	**10 High**	R02 R21	Process · Instrument
N8.5	Pressure	LESS	Loop pressure low	• Make-up supply lost or XV-105 closed • purge HIC-107 opened wide • PIC-105 transmitter reads high	Hydrogen partial pressure falls: conversion falls, benzene slips, product off-spec.	• PIC-105 LO alarm at 27.5 bar g • AI-103 HI alarm at 10 ppm	2	3	**6 Medium**	—	—
N8.6	Containment	OTHER THAN	Hydrogen escapes from the loop	• Flange or valve-stem leak • K-101 seal leak • small-bore fitting fatigue	Hydrogen jet fire or explosion near hot equipment; PIC-105 hides the loss by opening PV-105.	• PV-105 output and make-up FI-106 rise (hydrogen balance on the Hydrogen loop display) • EDV-101 depressuring to flare by the operator	5	2	**10 High**	R25	Safety / Instrument
N8.7	Composition	LESS	Hydrogen purity low (inerts build up)	• Methane rises in the make-up • purge reduced or blocked • nitrogen left on through HV-120	Hydrogen partial pressure falls: conversion falls and benzene slips to R-102 / R-103 (more heat there), product off-spec.	• AI-105 LO alarm at 85 % • purge HIC-107 • feed permissive purity ≥ 80 % (start only)	3	3	**9 Medium**	R23	Instrument / Operations
N8.8	Composition	AS WELL AS	Oxygen in the loop with hydrogen	• Air not purged out before hydrogen admission • air drawn in during maintenance • contaminated nitrogen	Flammable hydrogen–oxygen mixture inside equipment: internal explosion.	• AI-100 HI alarm at 1 % • XV-105 permissive AI-100 < 1 %	5	2	**10 High**	R26	Instrument / Operations
N8.9	Flow	REVERSE	Gas flows back through K-101	• K-101 stops with the discharge at 32.0 bar g	Reverse rotation, seal and bearing damage, hydrogen release at the seal.	• None shown (no non-return valve on the P&ID)	3	3	**9 Medium**	R27	Mechanical
N8.10	Temperature	MORE	K-101 discharge hot	• K-101 running on kickback with FV-103 closed • hot V-104 gas (see N7)	Compressor valves and seals damaged; hydrogen release.	• K-101 low forward-flow alarm	3	3	**9 Medium**	R27	Mechanical

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N8.11	Flow	MORE	Purge flow high	• HIC-107 opened wide	Hydrogen lost to fuel gas (cost); the fuel-gas system receives more hydrogen-rich gas.	• FI-107 indication • PIC-105 holds pressure	1	3	**3 Low**	—	—

A1.9 Node N9 · Nitrogen, vent, EDV-101 and flare

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N9 · Nitrogen, vent, EDV-101 and flare	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Purge the loop with nitrogen (8 bar g) through HV-120 before hydrogen and before maintenance; vent through HV-121 and depressure through EDV-101 to the flare.		
Design conditions	Loop oxygen AI-100 0.00 % · Loop pressure PIC-105 29.0 bar g (simulator design state)		
Equipment	HV-120, HV-121, EDV-101, flare header	Instruments	AI-100, PIC-105

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Flow	NO / NOT	No nitrogen for purging	• N2 header lost • HV-120 not opened	Oxygen cannot be removed before hydrogen admission (start-up held); equipment cannot be gas-freed for maintenance.	• XV-105 permissive AI-100 < 1 % • AI-100 HI alarm at 1 %	2	3	**6 Medium**	—	—
N9.2	Flow	REVERSE	Hydrogen into the nitrogen header	• HV-120 opened or passing while the loop is above the N2 header pressure (8 bar g)	Hydrogen in the plant nitrogen network: fire where the nitrogen is used for inerting.	• Nitrogen only flows while the loop is below the header pressure (procedure)	4	3	**12 High**	R28	Process / Mechanical
N9.3	Flow	NO / NOT	EDV-101 does not open on demand	• Actuator or solenoid failure • isolation valve left closed	Loop cannot be depressured in a fire or a leak: vessels weaken under fire exposure while at pressure.	• HV-121 vent to flare (manual)	5	2	**10 High**	R29	Process
N9.4	Flow	MORE	Vent and depressuring together exceed the flare capacity	• HV-121 and EDV-101 open while other units relieve	Flare header back-pressure: relief from other equipment restricted.	• None on the P&ID	3	2	**6 Medium**	R29	Process
N9.5	Flow	AS WELL AS	Liquid to the flare	• Depressuring with R-101 or V-104 level high • liquid swell on flashing	Burning liquid rain from the flare tip.	• LIC-101 HI alarm at 82 % • LIC-104 HI alarm at 78 %	4	2	**8 Medium**	R30	Process
N9.6	Composition	AS WELL AS	Air in the flare header	• Flare purge lost • header opened during maintenance	Flashback and explosion in the flare header when hydrogen is vented.	• None on the P&ID	4	2	**8 Medium**	R30	Process
N9.7	Pressure	LESS	Depressuring too fast	• EDV-101 and HV-121 open with the reactors hot	Hot liquid flashes: level swell, carry-over to E-103, R-102 and the flare; reactor contents boil.	• Any SIS trip stops benzene and hydrogen before depressuring	3	3	**9 Medium**	R29	Process
N9.8	Temperature	LESS	Metal chilled by flashing liquid during depressuring	• Liquid auto-refrigerates as the pressure falls (cold, liquid-full equipment)	Carbon steel below its minimum design metal temperature: brittle fracture on repressuring.	• None on the P&ID	4	1	**4 Low**	R29	Process

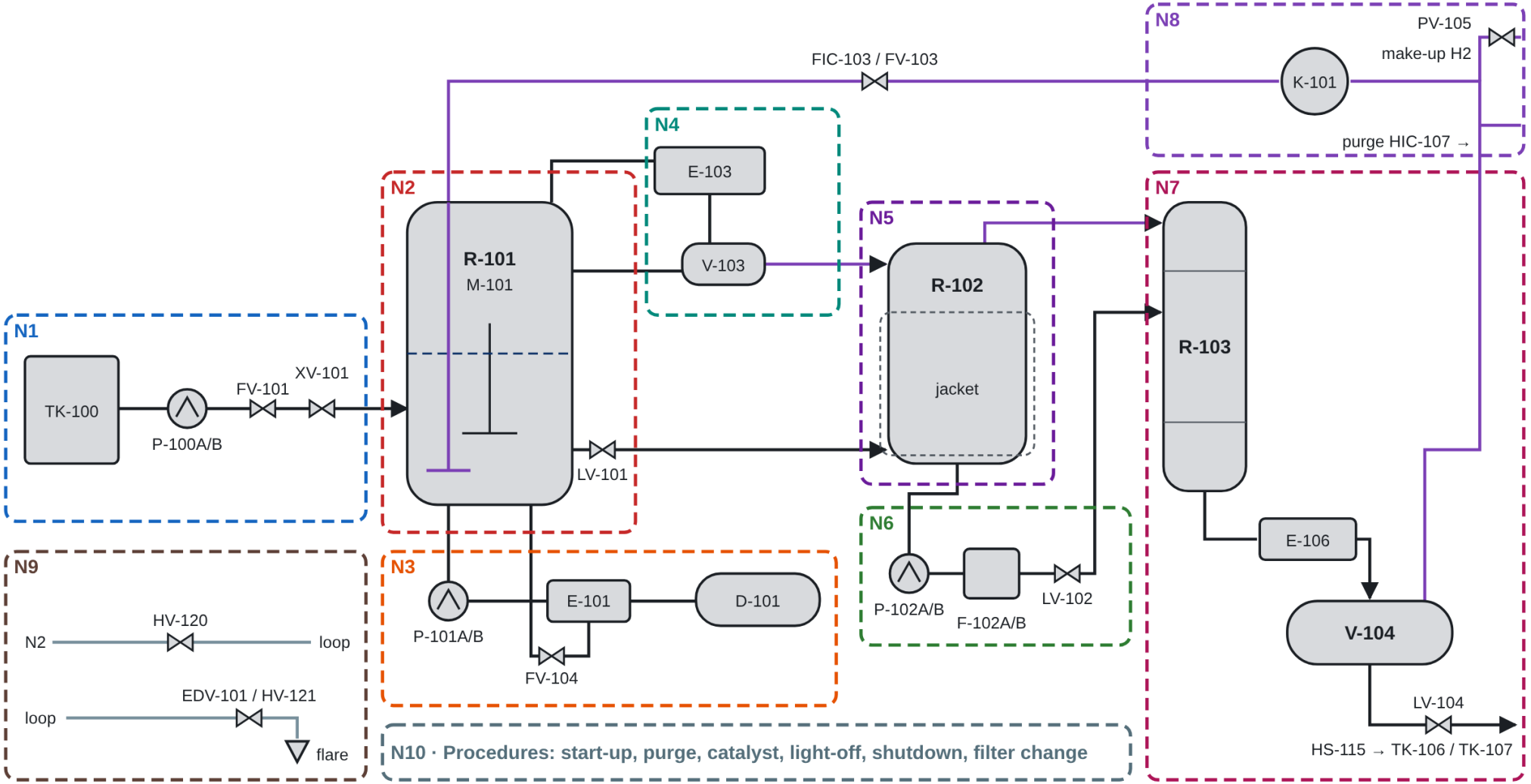
A1.10 Node N10 · Procedures

Project	Cyclohexane Reactor Ver 1 · operator training simulator (teaching HAZOP)	Drawing	CHX-V1-PID-001 · CHX-V1-PFD-001
Node	N10 · Procedures	Study	CHX-V1-SA-HZP-001 rev A
Design intent	Start up (nitrogen purge to below 1 % O2, hydrogen admission, catalyst charge, heat-up on MP steam, light-off at TI-100 ≥ 140 °C and feed ramp), shut down, and change over the catalyst filter without hazard to people or equipment.		
Design conditions	(simulator design state)		
Equipment	Whole unit	Instruments	SIS permissives and resets, HS-101, HS-102, HS-115, HV-110, HV-117, HV-120, HV-121, F-102 A/B

ID	Parameter	Guide word	Deviation	Causes	Consequences (no safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N10.1	Sequence	BEFORE	Hydrogen admitted before the oxygen is purged	• Purge step shortened • AI-100 sample point not representative	Hydrogen–air mixture inside R-101, R-102 and the loop: internal explosion.	• XV-105 permissive AI-100 < 1 % • AI-100 HI alarm at 1 %	5	1	**5 Medium**	R26	Instrument / Operations
N10.2	Sequence	BEFORE	Benzene admitted before the reactor is ready	• Feed started before hydrogen, agitation, level or light-off temperature are established	Benzene accumulates without reacting, then reacts all at once: temperature runaway.	• XV-101 permissives: SIS reset, AI-105 ≥ 80 %, LIC-101 ≥ 40 %, M-101 running, K-101 running with FIC-103 ≥ 15 000 Nm³/h, TI-100 ≥ 140 °C, HV-113 open	4	1	**4 Low**	—	—
N10.3	Time	EARLY	Feed ramped up too fast after light-off	• Benzene raised to full rate before the pump-around and condenser carry the load	Heat release outruns heat removal: TIC-101 overshoots toward the trip.	• TIC-101 HI alarm at 208 °C • TSHH-100 trips at 220 °C • XV-101 alarm with R-101 below 180 °C	4	3	**12 High**	R32	Operations / Instrument
N10.4	Time	LATE	HS-101 switched from HEAT to COOL too late	• Operator busy with the feed ramp • no step for it in the light-off sequence	MP steam still heats E-101 while the reaction starts: rapid temperature rise.	• TIC-101 HI alarm at 208 °C • TSHH-100 sets HS-101 to COOL and closes HV-110	4	3	**12 High**	R16	Instrument / Operations
N10.5	Sequence	AFTER	Catalyst charged after the feed is at full rate	• Catalyst top-up through HV-117 while reacting at design rate	Activity step raises the heat release: temperature overshoot.	• TIC-101 cascade • TIC-101 HI alarm at 208 °C	3	2	**6 Medium**	R31	Operations / Safety
N10.6	Maintenance	OTHER THAN	Dry Raney-Ni exposed to air during catalyst handling	• Catalyst allowed to dry in the pot, a drum or the filter cake	Pyrophoric catalyst ignites: fire, burns, ignition of hydrocarbon nearby.	• Catalyst pot HV-117 charging under liquid (procedure)	3	3	**9 Medium**	R31	Operations / Safety
N10.7	Sequence	PART OF	Shutdown purge incomplete before equipment is opened	• Gas-freeing steps skipped or not verified (hydrogen, cyclohexane left inside)	Flammable release or fire when the flange is broken; pyrophoric catalyst exposed to air.	• AI-100 and AI-105 cover the loop only, not the item opened	4	3	**12 High**	R34	Operations
N10.8	Sequence	REVERSE	Shutdown steps in reverse order: hydrogen stopped before the benzene	• K-101 stopped or FV-103 closed while XV-101 is still open	Benzene left unreacted in R-101 and R-102 with catalyst; heat released at the next start.	• XS-102 trips the unit when K-101 stops with XV-101 open (10 s) • FIC-103 LO alarm at 35000 Nm³/h	3	2	**6 Medium**	R32	Operations / Instrument
N10.9	Sequence	OTHER THAN	Product lined up to the on-spec tank during start-up	• HS-115 left on TK-106 while the product is still off-spec	On-spec tank contaminated with benzene.	• AI-103 HI alarm at 10 ppm • HS-115 to TK-107	2	3	**6 Medium**	R18	Instrument / Operations
N10.10	Time	LATE	Filter change-over late	• PDI-102 alarm not acted on	R-102 level rises toward the gas outlet (see N5).	• PDI-102 HI alarm at 0.8 bar • LIC-102 HI alarm at 80 %	2	3	**6 Medium**	R34	Operations
N10.11	Time	EARLY	SIS reset before the cause is cleared	• Reset attempted while a trip condition is still active or R-101 is still hot	Restart into the same upset.	• SIS reset is blocked while a trip condition is active or R-101 is above the reset temperature	3	1	**3 Low**	—	—

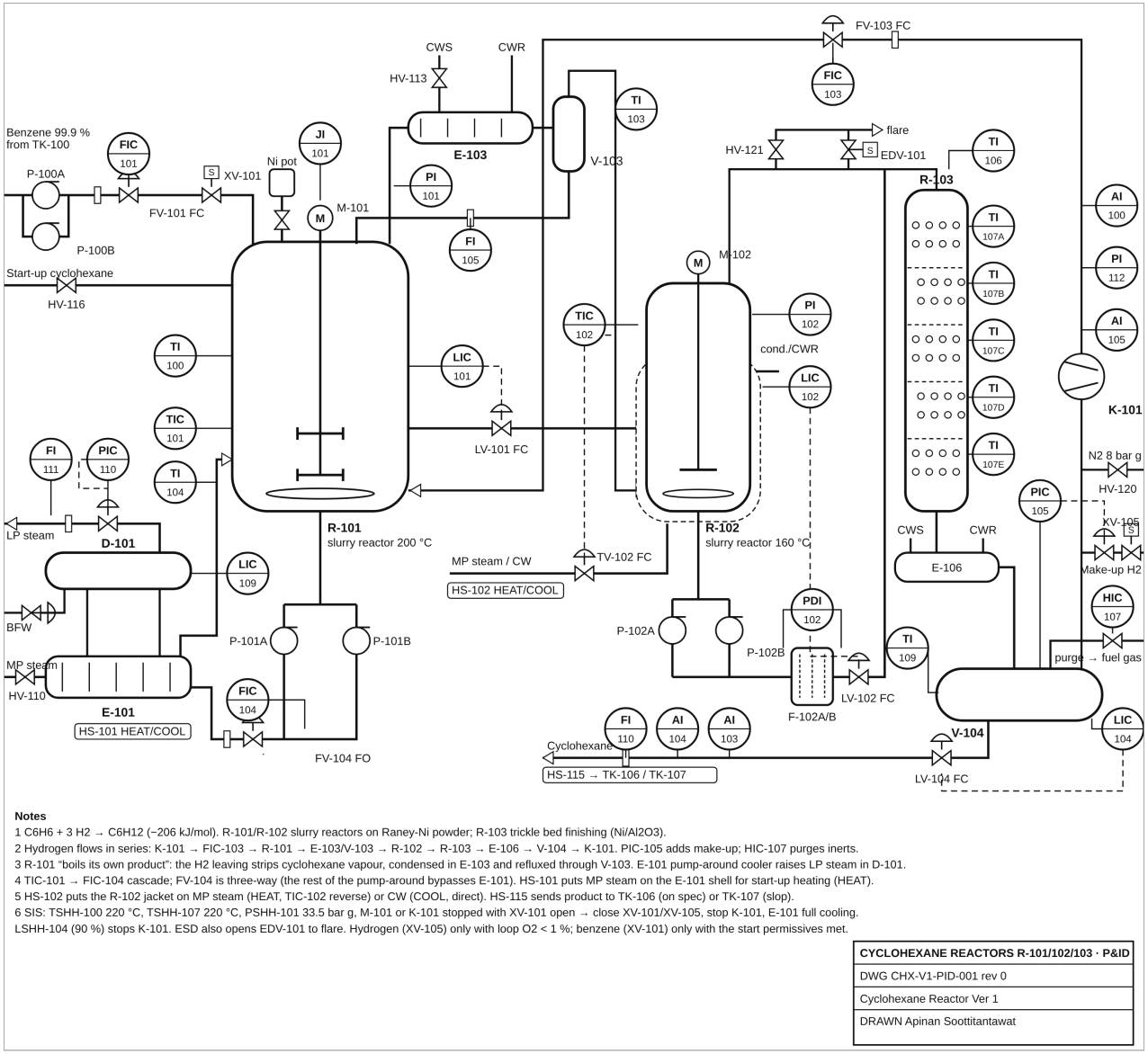
Appendix 2 Drawings

A2.1 HAZOP node map



Rendered from the HAZOP tab of the simulator (hzMapSvg); nodes N1–N10 as in section 2.

A2.2 Screen P&ID CHX-V1-PID-001

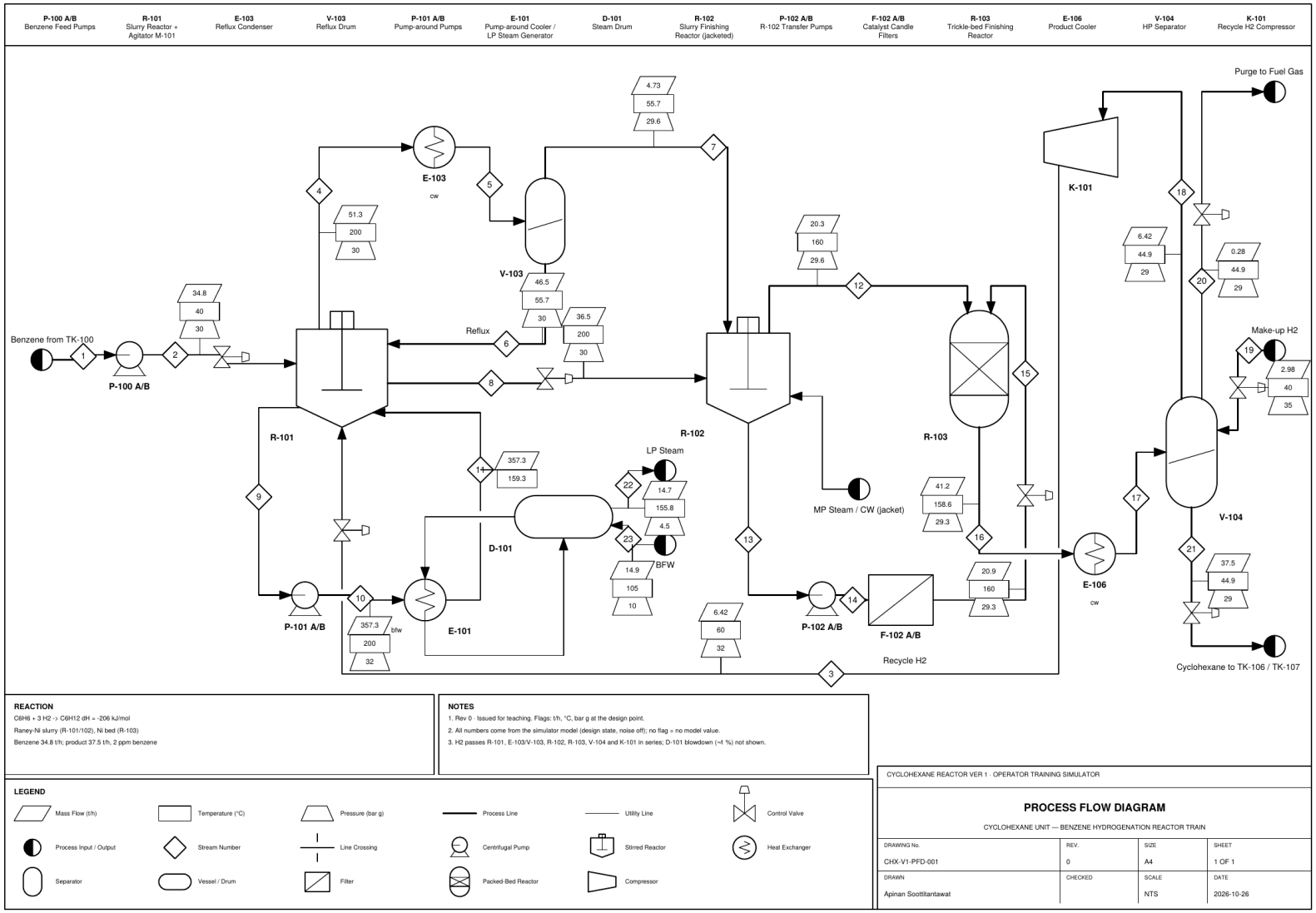


Notes

1 C6H6 + 3 H2 → C6H12 (−206 kJ/mol). R-101/R-102 slurry reactors on Raney-Ni powder; R-103 trickle bed finishing (Ni/Al2O3).
2 Hydrogen flows in series: K-101 → FIC-103 → R-101 → E-103/V-103 → R-102 → R-103 → E-106 → V-104 → K-101. PIC-105 adds make-up; HIC-107 purges inerts.
3 R-101 "boils its own product": the H2 leaving strips cyclohexane vapour, condensed in E-103 and refluxed through V-103. E-101 pump-around cooler raises LP steam in D-101.
4 TIC-101 → FIC-104 cascade; FV-104 is three-way (the rest of the pump-around bypasses E-101). HS-101 puts MP steam on the E-101 shell for start-up heating (HEAT).
5 HS-102 puts the R-102 jacket on MP steam (HEAT, TIC-102 reverse) or CW (COOL, direct). HS-115 sends product to TK-106 (on spec) or TK-107 (slop).
6 SIS: TSHH-100 220 °C, TSHH-107 220 °C, PSHH-101 33.5 bar g, M-101 or K-101 stopped with XV-101 open → close XV-101/XV-105, stop K-101, E-101 full cooling.
LSHH-104 (90 %) stops K-101. ESD also opens EDV-101 to flare. Hydrogen (XV-105) only with loop O2 < 1 %; benzene (XV-101) only with the start permissives met.

CYCLOHEXANE REACTORS R-101/102/103 · P&ID
DWG CHX-V1-PID-001 rev 0
Cyclohexane Reactor Ver 1
DRAWN Apinan Soottitantawat

A2.3 Process flow diagram CHX-V1-PFD-001



Appendix 3 Action close-out sheets

One sheet per recommendation (38 sheets). The action party writes the response; the HAZOP leader ticks the status and signs. Names, signatures and dates are blank.

Action close-out sheet R01

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R01**
Node(s)	N1 Benzene feed, N7 R-103, E-106, V-104 and product	Action party	Instrument / Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.2	Flow MORE: More benzene than the hydrogen and heat removal can take	FIC-101 output driven high (controller or positioner fault, set-point entry error); FI-101 transmitter reading low; FV-101 trim damaged	Conversion in R-101 falls and unreacted benzene passes to R-102 and R-103, which then release its heat of reaction: R-102 and the adiabatic R-103 bed heat up (about 9 °C per 1 % benzene slip), the product goes off-spec and the bed can run away.	FIC-101 HI alarm at 42 t/h; TIC-102 HI alarm at 172 °C; TI-107 HI alarm at 185 °C; TSHH-107 trips at 220 °C (5 s); TSHH-100 trips at 220 °C (5 s); AI-103 HI alarm at 10 ppm	4×3 = 12 High
N7.1	Temperature MORE: R-103 bed hotter than design (hot spot)	Benzene slip from R-102 (each 1 % benzene gives about 9 °C adiabatic rise); hot inlet from R-102	Bed runaway: catalyst sintered, isomer formed, vessel wall above design temperature, loss of containment.	TI-107 HI alarm at 185 °C (hottest of five); TSHH-107 trips at 220 °C (5 s); AI-104 HI alarm at 200 ppm	4×3 = 12 High

Recommendation

Add a high-high benzene flow or benzene-to-hydrogen ratio trip (FI-101 against FI-103) that closes XV-101; the feed can run far above design with no trip.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R02

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R02**
Node(s)	N2 R-101 slurry reactor, N4 Reflux condenser E-103 / V-103, N8 Hydrogen loop	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.1	Temperature MORE: R-101 above 200.0 °C toward its boiling point (about 258 °C at reactor pressure)	E-101 heat removal lost or reduced (see N3); reflux condenser lost (see N4); TIC-101 measurement reading low; TIC-101 left in MAN; too much catalyst charged; feed rate too high	Faster reaction and more isomer (MCP); at the boiling ceiling the extra heat becomes vapour, the pressure and the condenser load rise and hot hydrocarbon is carried to R-102. Overpressure of R-101 if the vapour cannot leave.	TIC-101 HI alarm at 208 °C; TI-100 independent element HI alarm at 210 °C; TSHH-100 trips at 220 °C (5 s): XV-101 and XV-105 shut, K-101 stops, E-101 to COOL with FV-104 fully open; PSHH-101 trips at 33.5 bar g	4×3 = 12 High
N2.7	Pressure MORE: R-101 pressure above normal	Make-up valve PV-105 driven open (PIC-105 output or transmitter fault); restriction in the series gas path R-101 → E-103 / V-103 → R-102 → R-103 → V-104; boiling at the ceiling	Overpressure of R-101 and the loop: loss of containment of hydrogen and hot hydrocarbon, fire or explosion.	PIC-105 HI alarm at 30.5 bar g (measured at V-104); PSHH-101 trips at 33.5 bar g (3 s): XV-105 make-up and XV-101 shut, K-101 stops	5×2 = 10 High
N4.5	Pressure MORE: R-101 pressure rises against a restricted vent path	Vent line V-103 → R-102 fouled or a valve in it partly closed	R-101 pressure rises while PIC-105, which measures at V-104, sees nothing: R-101 relies on PSHH-101 alone.	PSHH-101 trips at 33.5 bar g (3 s); PI-101 indication (no alarm)	4×2 = 8 Medium
N8.4	Pressure MORE: Loop pressure high	PV-105 driven open (PIC-105 output high or its transmitter reads low); make-up supply pressure high	Loop overpressured from the make-up header: loss of containment of hydrogen, fire or explosion.	PIC-105 HI alarm at 30.5 bar g (same transmitter as the controller); PSHH-101 trips at 33.5 bar g (3 s) and closes XV-105	5×2 = 10 High

Recommendation

Confirm and document the pressure relief of R-101, R-102, R-103 and V-104 for blocked outlet, make-up valve wide open and fire; relief valves are not shown on the training P&ID.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R03

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R03**
Node(s)	N1 Benzene feed, N2 R-101 slurry reactor, N4 Reflux condenser E-103 / V-103, N5 R-102 and its jacket	Action party	Instrument / Safety
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.2	Flow MORE: More benzene than the hydrogen and heat removal can take	FIC-101 output driven high (controller or positioner fault, set-point entry error); FI-101 transmitter reading low; FV-101 trim damaged	Conversion in R-101 falls and unreacted benzene passes to R-102 and R-103, which then release its heat of reaction: R-102 and the adiabatic R-103 bed heat up (about 9 °C per 1 % benzene slip), the product goes off-spec and the bed can run away.	FIC-101 HI alarm at 42 t/h; TIC-102 HI alarm at 172 °C; TI-107 HI alarm at 185 °C; TSHH-107 trips at 220 °C (5 s); TSHH-100 trips at 220 °C (5 s); AI-103 HI alarm at 10 ppm	4×3 = 12 High
N2.4	Mixing LESS: Impeller turns slowly while the motor runs	Coupling or shaft failure; impeller loose or lost	As for no agitation, but the motor-run signal stays healthy so XS-101 does not act: benzene builds up in R-101 and R-102 receives it; R-102 has no high-high temperature trip.	JI-101 LO alarm at 80 kW; TIC-101 LO alarm at 190 °C; TIC-102 HI alarm at 172 °C	4×3 = 12 High
N4.1	Flow NO / NOT: No cooling water to E-103	CW supply lost (pump trip at the cooling tower); HV-113 closed in error	Reflux FI-105 stops: R-101 loses a heat sink and its level falls; hot cyclohexane vapour goes with the hydrogen to R-102, which heats, and on to R-103.	TI-103 HI alarm at 75 °C; LIC-101 LO alarm at 45 %; TIC-102 HI alarm at 172 °C; TSHH-107 trips at 220 °C; feed permissive HV-113 open (start only)	4×3 = 12 High
N5.1	Temperature MORE: R-102 hotter than 160.0 °C	Benzene slip from R-101 (more feed, low activity or poor mixing there); hot vapour from E-103 when its cooling is lost; HS-102 left on HEAT when cooling is needed; TV-102 output high in HEAT	More vapour and benzene go on to R-103, whose adiabatic bed heats; R-102 approaches its boiling ceiling and has no high-high trip of its own.	TIC-102 HI alarm at 172 °C; TV-102 fails closed (no steam in HEAT); any SIS trip sets HS-102 to COOL; TSHH-107 trips at 220 °C	4×3 = 12 High

Recommendation

Add a high-high temperature trip on R-102 with its own element (set below the R-102 boiling point at loop pressure).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R04

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R04**
Node(s)	N2 R-101 slurry reactor	Action party	Instrument / Safety
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.4	Mixing LESS: Impeller turns slowly while the motor runs	Coupling or shaft failure; impeller loose or lost	As for no agitation, but the motor-run signal stays healthy so XS-101 does not act: benzene builds up in R-101 and R-102 receives it; R-102 has no high-high temperature trip.	JI-101 LO alarm at 80 kW; TIC-101 LO alarm at 190 °C; TIC-102 HI alarm at 172 °C	4×3 = 12 High

Recommendation

Add low-low agitator power (JI-101) to the SIS so that a running motor with a failed drive is seen like a stopped motor.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R05

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R05**
Node(s)	N2 R-101 slurry reactor	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.5	Flow NO / NOT: No overflow from R-101 to R-102	LV-101 stuck or failed closed (loss of instrument air); LIC-101 transmitter reads low; LIC-101 left in MAN	R-101 level rises: less vapour space, liquid and foam carried into the vapour line to E-103 / V-103 and with the gas to R-102; downstream reactors starve.	LIC-101 HI alarm at 82 %; LIC-101 in AUTO	3×3 = 9 Medium
N2.6	Level LESS: R-101 level low	LIC-101 output high or transmitter reads high; boil-off not returned (reflux lost, see N4); feed lost with LV-101 open	Sparger and agitator uncovered (start permissive is 40 %), pump-around P-101A/B loses suction and cavitates; less liquid to absorb heat.	LIC-101 LO alarm at 45 %; level permissive ≥ 40 % before feed (start only)	3×3 = 9 Medium

Recommendation

Add an independent R-101 level element with high-high and low-low actions (stop feed; protect P-101A/B).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R06

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R06**
Node(s)	N4 Reflux condenser E-103 / V-103	Action party	Process / Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N4.1	Flow NO / NOT: No cooling water to E-103	CW supply lost (pump trip at the cooling tower); HV-113 closed in error	Reflux FI-105 stops: R-101 loses a heat sink and its level falls; hot cyclohexane vapour goes with the hydrogen to R-102, which heats, and on to R-103.	TI-103 HI alarm at 75 °C; LIC-101 LO alarm at 45 %; TIC-102 HI alarm at 172 °C; TSHH-107 trips at 220 °C; feed permissive HV-113 open (start only)	4×3 = 12 High
N4.4	Flow MORE: Vapour load above condenser capacity	R-101 at its boiling ceiling (heat removal lost elsewhere); hydrogen flow raised	Uncondensed vapour passes to R-102: R-102 and R-103 heat.	TI-103 HI alarm at 75 °C; TIC-101 HI alarm at 208 °C; TSHH-100 trips at 220 °C	3×3 = 9 Medium

Recommendation

Add low cooling-water flow on E-103 or high-high TI-103 to stop the benzene feed.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R07

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R07**
Node(s)	N5 R-102 and its jacket	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N5.3	Level MORE: R-102 level high	LV-102 closed or failed closed; P-102A stops; F-102 blinded; LIC-102 transmitter reads low	Vapour space lost and liquid carried with the gas to R-103 (slug on the bed); R-101 overflow keeps filling R-102.	LIC-102 HI alarm at 80 %; PDI-102 HI alarm at 0.8 bar	3×3 = 9 Medium
N5.4	Level LESS: R-102 level low	LIC-102 transmitter reads high; LV-102 output high	P-102A/B run dry or cavitate; gas blows through the filter into the R-103 liquid inlet.	LIC-102 LO alarm at 30 %	2×3 = 6 Medium

Recommendation

Add independent high-high and low-low level on R-102 (stop P-102A/B, stop feed).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R08

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R08**
Node(s)	N1 Benzene feed	Action party	Process / Mechanical
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	10 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.4	Flow REVERSE: Reactor liquid and hydrogen flow back toward P-100A/B and TK-100	Feed pump stops while R-101 is at 30.0 bar g with FV-101 and XV-101 open; no non-return valve shown on the pump discharge	Hot liquid saturated with hydrogen enters the feed line and the atmospheric tank TK-100: tank overpressure, release of hydrogen and benzene vapour, fire.	XV-101 closes on any SIS trip; FV-101 fails closed; FIC-101 LO alarm at 25 t/h	5×2 = 10 High

Recommendation

Fit a non-return valve on the P-100A/B discharge and close XV-101 when no feed pump runs.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R09

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R09**
Node(s)	N1 Benzene feed	Action party	Process / Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.7	Composition OTHER THAN: Sulphur compounds (thiophene) in the benzene	Off-spec benzene cargo; TK-100 contaminated; wrong tank lined up	The Raney-Ni in R-101 is poisoned irreversibly: conversion falls, benzene slips to R-102 and R-103 (more heat there), product off-spec; the catalyst must be replaced.	AI-103 HI alarm at 10 ppm; TI-107 HI alarm at 185 °C; fresh catalyst through the catalyst pot HV-117	3×3 = 9 Medium
N1.8	Composition AS WELL AS: Water with the benzene	Tank bottoms drawn into the suction; water not drained from TK-100	Water flashes in R-101 at 200 °C: pressure surge, catalyst activity lost, water carried to V-104 and the product.	TIC-101 and PIC-105 control; PSHH-101 trips at 33.5 bar g (3 s)	2×3 = 6 Medium

Recommendation

Analyse every benzene receipt for sulphur and water; drain TK-100 water; consider a guard bed ahead of R-101.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R10

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R10**
Node(s)	N1 Benzene feed, N3 Pump-around, E-101 and steam drum, N6 Transfer and catalyst filter	Action party	Electrical / Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	8 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.1	Flow NO / NOT: No benzene flow to R-101	Running feed pump P-100A stops (motor trip) and P-100B is not started; FV-101 closes (fails closed on loss of instrument air); XV-101 closed by the SIS or spuriously; TK-100 empty or suction strainer blocked	Reaction stops: R-101 cools, hydrogen consumption falls and the loop pressure rises until PIC-105 closes PV-105; production lost and a new light-off needed. A pump running against a closed FV-101 or XV-101 overheats and its seal can leak benzene.	FIC-101 LO alarm at 25 t/h; standby pump P-100B; TIC-101 LO alarm at 190 °C; PIC-105 holds loop pressure by closing PV-105	2×3 = 6 Medium
N3.1	Flow NO / NOT: No pump-around flow through E-101	P-101A stops (motor trip) and P-101B is not started; R-101 level below the pump suction; FIC-104 output driven to 0 % (controller fault)	E-101 removes no heat (design 9.6 MW): R-101 heats toward the trip and the boiling ceiling; D-101 steam make falls and LP steam export stops.	TIC-101 HI alarm at 208 °C; TI-100 HI alarm at 210 °C; TSHH-100 trips at 220 °C (5 s); FV-104 fails open; standby pump P-101B	4×2 = 8 Medium
N6.1	Flow NO / NOT: No liquid from R-102 to R-103	P-102A stops (motor trip) and P-102B is not started; LV-102 fails closed (loss of instrument air); F-102 fully blinded	R-102 fills (see N5); R-103 receives vapour only and the product rate to V-104 falls.	LIC-102 HI alarm at 80 %; LIC-104 LO alarm at 25 %; standby pump P-102B	2×3 = 6 Medium

Recommendation

Auto-start the standby pumps P-100B, P-101B and P-102B on low discharge pressure or flow.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R11

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R11**
Node(s)	N3 Pump-around, E-101 and steam drum	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	8 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N3.1	Flow NO / NOT: No pump-around flow through E-101	P-101A stops (motor trip) and P-101B is not started; R-101 level below the pump suction; FIC-104 output driven to 0 % (controller fault)	E-101 removes no heat (design 9.6 MW): R-101 heats toward the trip and the boiling ceiling; D-101 steam make falls and LP steam export stops.	TIC-101 HI alarm at 208 °C; TI-100 HI alarm at 210 °C; TSHH-100 trips at 220 °C (5 s); FV-104 fails open; standby pump P-101B	4×2 = 8 Medium

Recommendation

Add a low-flow alarm on the pump-around FI-104 (none in the alarm list).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R12

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R12**
Node(s)	N3 Pump-around, E-101 and steam drum	Action party	Process / Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N3.3	Temperature MORE: Pump-around return TI-104 hotter than design	E-101 fouled by polymer and catalyst fines; low water level in D-101 uncovering tubes	Heat-transfer margin used up slowly while TIC-101 hides it by opening FV-104; once FV-104 is wide open R-101 temperature rises and the unit trips or must cut rate.	TIC-101 HI alarm at 208 °C; FIC-104 output shown on the faceplate; TSHH-100 trips at 220 °C	3×3 = 9 Medium

Recommendation

Add a control-margin alarm on the FIC-104 output and trend the E-101 heat-transfer coefficient.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R13

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R13**
Node(s)	N3 Pump-around, E-101 and steam drum	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N3.4	Pressure MORE: D-101 pressure above 4.50 bar g	High LP steam header pressure (a large user trips) so PV-110 cannot export; PV-110 stuck closed; PIC-110 transmitter reads low	Higher saturation temperature in D-101 lowers E-101 duty: R-101 heats; drum overpressure if export stays blocked.	PIC-110 HI alarm at 5.6 bar g; PV-110 fails open; TIC-101 HI alarm at 208 °C; TSHH-100 trips at 220 °C	3×3 = 9 Medium

Recommendation

Confirm the D-101 steam-side relief for blocked export at maximum E-101 duty.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R14

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R14**
Node(s)	N3 Pump-around, E-101 and steam drum	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N3.6	Level LESS: D-101 water level low	Boiler feed water lost; LV-109 closed in MAN; LIC-109 transmitter reads high	E-101 tubes uncovered: heat removal falls (R-101 heats) and refilling a dry hot drum causes thermal shock.	LIC-109 LO alarm at 25 %; LV-109 fails open	3×3 = 9 Medium
N3.7	Level MORE: D-101 water level high	LV-109 fails open (loss of instrument air); LIC-109 transmitter reads low	Water carried into the LP steam header: water hammer in steam users.	LIC-109 HI alarm at 75 %	3×3 = 9 Medium

Recommendation

Add low-low and high-high level on D-101 independent of LIC-109.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R15

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R15**
Node(s)	N3 Pump-around, E-101 and steam drum	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	8 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N3.8	Composition AS WELL AS: Hydrocarbon and hydrogen in the steam system	E-101 tube leak: the process side (30.0 bar g) is above the steam side (4.50 bar g)	Cyclohexane, benzene and hydrogen enter D-101 and the LP steam header: flammable release at steam users and vents.	None on the P&ID	4×2 = 8 Medium

Recommendation

Monitor D-101 blowdown or condensate for hydrocarbon (E-101 tube leak).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R16

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R16**
Node(s)	N3 Pump-around, E-101 and steam drum, N10 Procedures	Action party	Instrument / Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N3.9	Utility OTHER THAN: MP steam on E-101 while reacting (HS-101 on HEAT)	HS-101 not switched from HEAT to COOL after light-off; HV-110 left open	E-101 heats R-101 instead of cooling it: temperature rises quickly at full rate.	TIC-101 HI alarm at 208 °C; TSHH-100 trips at 220 °C and sets HS-101 to COOL with HV-110 closed	4×2 = 8 Medium
N10.4	Time LATE: HS-101 switched from HEAT to COOL too late	Operator busy with the feed ramp; no step for it in the light-off sequence	MP steam still heats E-101 while the reaction starts: rapid temperature rise.	TIC-101 HI alarm at 208 °C; TSHH-100 sets HS-101 to COOL and closes HV-110	4×3 = 12 High

Recommendation

Interlock HV-110 (MP steam to E-101) closed whenever XV-101 is open, and put the HEAT to COOL switch-over in the light-off steps.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R17

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R17**
Node(s)	N4 Reflux condenser E-103 / V-103, N5 R-102 and its jacket	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	8 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N4.6	Composition AS WELL AS: Hydrogen and hydrocarbon in the cooling water	E-103 tube leak: process side (30.0 bar g) above the CW side	Hydrogen and cyclohexane carried to the cooling tower: flammable cloud at the tower, fire.	None on the P&ID	4×2 = 8 Medium
N5.6	Composition AS WELL AS: Hydrocarbon in the jacket medium	Jacket wall leak: process (29.6 bar g) above steam (10 bar g) or CW	Hydrogen and hydrocarbon into condensate or cooling water: flammable release at the tower or the condensate tank.	None on the P&ID	3×2 = 6 Medium

Recommendation

Detect hydrogen and hydrocarbon in the CW return of E-103, E-106 and the R-102 jacket (degassing pot or analyser).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R18

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R18**
Node(s)	N2 R-101 slurry reactor, N5 R-102 and its jacket, N7 R-103, E-106, V-104 and product, N10 Procedures	Action party	Instrument / Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	6 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.10	Composition AS WELL AS: Isomer (MCP) formed with the cyclohexane	R-101 running hot; R-103 bed hot	Product off-spec on MCP (spec 200 ppm) sent to the on-spec tank.	AI-104 HI alarm at 200 ppm; HS-115 routes product to TK-107	1×4 = 4 Low
N5.2	Flow NO / NOT: No heating medium in the R-102 jacket	MP steam lost; TV-102 fails closed (loss of instrument air); HS-102 on COOL in error	R-102 cools: conversion there falls and more benzene reaches R-103; product can go off-spec.	TIC-102 LO alarm at 148 °C; R-103 finishes the benzene; AI-103 HI alarm at 10 ppm	2×3 = 6 Medium
N7.7	Composition OTHER THAN: Off-spec product sent to the on-spec tank	Benzene slip; analyser AI-103 frozen on an old value; HS-115 not switched to TK-107	TK-106 contaminated with benzene above 10 ppm: tank downgraded, customer claim.	AI-103 HI alarm at 10 ppm; AI-104 HI alarm at 200 ppm; HS-115 to TK-107	2×3 = 6 Medium
N10.9	Sequence OTHER THAN: Product lined up to the on-spec tank during start-up	HS-115 left on TK-106 while the product is still off-spec	On-spec tank contaminated with benzene.	AI-103 HI alarm at 10 ppm; HS-115 to TK-107	2×3 = 6 Medium

Recommendation

Route product to TK-107 automatically on AI-103 or AI-104 high and alarm a stale analyser value.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R19

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R19**
Node(s)	N7 R-103, E-106, V-104 and product	Action party	Process / Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	10 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N7.6	Level LESS: V-104 level lost: gas blow-by to storage	LIC-104 transmitter reads high; LV-104 output high; LIC-104 in MAN with LV-104 open	Hydrogen at 29.0 bar g flows through LV-104 into the atmospheric tank TK-106 or TK-107: tank overpressure, rupture, fire.	LIC-104 LO alarm at 25 % (same transmitter as the controller); LV-104 fails closed	5×2 = 10 High

Recommendation

Add an independent low-low level on V-104 closing a shutdown valve in the product line; check the TK-106 / TK-107 vents for gas blow-by.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R20

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R20**
Node(s)	N6 Transfer and catalyst filter	Action party	Instrument / Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	6 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N6.4	Composition AS WELL AS: Catalyst fines with the liquid to R-103	Torn filter candle or gasket bypass on F-102	Raney-Ni fines foul the R-103 bed and reach V-104 and the product tanks (pyrophoric solids in storage).	PDI-102 indication (a torn candle shows a LOW ΔP, which has no alarm)	3×2 = 6 Medium

Recommendation

Add a low ΔP alarm on PDI-102 (torn candle) and a routine fines check of the product.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R21

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R21**
Node(s)	N2 R-101 slurry reactor, N4 Reflux condenser E-103 / V-103, N8 Hydrogen loop	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	10 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.7	Pressure MORE: R-101 pressure above normal	Make-up valve PV-105 driven open (PIC-105 output or transmitter fault); restriction in the series gas path R-101 → E-103 / V-103 → R-102 → R-103 → V-104; boiling at the ceiling	Overpressure of R-101 and the loop: loss of containment of hydrogen and hot hydrocarbon, fire or explosion.	PIC-105 HI alarm at 30.5 bar g (measured at V-104); PSHH-101 trips at 33.5 bar g (3 s): XV-105 make-up and XV-101 shut, K-101 stops	5×2 = 10 High
N4.5	Pressure MORE: R-101 pressure rises against a restricted vent path	Vent line V-103 → R-102 fouled or a valve in it partly closed	R-101 pressure rises while PIC-105, which measures at V-104, sees nothing: R-101 relies on PSHH-101 alone.	PSHH-101 trips at 33.5 bar g (3 s); PI-101 indication (no alarm)	4×2 = 8 Medium
N8.4	Pressure MORE: Loop pressure high	PV-105 driven open (PIC-105 output high or its transmitter reads low); make-up supply pressure high	Loop overpressured from the make-up header: loss of containment of hydrogen, fire or explosion.	PIC-105 HI alarm at 30.5 bar g (same transmitter as the controller); PSHH-101 trips at 33.5 bar g (3 s) and closes XV-105	5×2 = 10 High

Recommendation

Add a PI-101 high alarm between the PIC-105 alarm and PSHH-101, because PIC-105 measures at V-104, not at R-101.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R22

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R22**
Node(s)	N7 R-103, E-106, V-104 and product	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N7.3	Temperature MORE: V-104 separator hot	E-106 cooling water lost or reduced; warm CW	More cyclohexane vapour to K-101 and the purge; hotter compressor suction; product flashes in storage.	TI-109 indication (no alarm)	3×3 = 9 Medium

Recommendation

Add a high alarm on TI-109 (V-104 temperature).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R23

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R23**
Node(s)	N8 Hydrogen loop	Action party	Instrument / Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N8.7	Composition LESS: Hydrogen purity low (inerts build up)	Methane rises in the make-up; purge reduced or blocked; nitrogen left on through HV-120	Hydrogen partial pressure falls: conversion falls and benzene slips to R-102 / R-103 (more heat there), product off-spec.	AI-105 LO alarm at 85 %; purge HIC-107; feed permissive purity ≥ 80 % (start only)	3×3 = 9 Medium

Recommendation

Add a low-flow alarm on the purge FI-107 and a purity check in the shift routine.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R24

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R24**
Node(s)	N8 Hydrogen loop	Action party	Instrument / Safety
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N8.2	Flow LESS: Hydrogen flow low with K-101 running	FV-103 fails closed (loss of instrument air) or FIC-103 output low; FI-103 transmitter reads high	Same as no hydrogen in R-101, but the compressor keeps running so XS-102 does not act: benzene accumulates and passes on to R-102 and R-103; K-101 runs on kickback.	FIC-103 LO alarm at 35000 Nm³/h; K-101 low forward-flow alarm; TIC-101 LO alarm at 190 °C; AI-103 HI alarm at 10 ppm	4×3 = 12 High

Recommendation

Add low-low hydrogen flow (FI-103) to the SIS to close XV-101; today only a stopped K-101 motor is seen.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R25

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R25**
Node(s)	N8 Hydrogen loop	Action party	Safety / Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	10 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N8.6	Containment OTHER THAN: Hydrogen escapes from the loop	Flange or valve-stem leak; K-101 seal leak; small-bore fitting fatigue	Hydrogen jet fire or explosion near hot equipment; PIC-105 hides the loss by opening PV-105.	PV-105 output and make-up FI-106 rise (hydrogen balance on the Hydrogen loop display); EDV-101 depressuring to flare by the operator	5×2 = 10 High

Recommendation

Install hydrogen gas detection at K-101, the reactors and V-104, and alarm the loop hydrogen balance (make-up against 3 × benzene fed).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R26

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R26**
Node(s)	N8 Hydrogen loop, N10 Procedures	Action party	Instrument / Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	10 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N8.8	Composition AS WELL AS: Oxygen in the loop with hydrogen	Air not purged out before hydrogen admission; air drawn in during maintenance; contaminated nitrogen	Flammable hydrogen–oxygen mixture inside equipment: internal explosion.	AI-100 HI alarm at 1 %; XV-105 permissive AI-100 < 1 %	5×2 = 10 High
N10.1	Sequence BEFORE: Hydrogen admitted before the oxygen is purged	Purge step shortened; AI-100 sample point not representative	Hydrogen–air mixture inside R-101, R-102 and the loop: internal explosion.	XV-105 permissive AI-100 < 1 %; AI-100 HI alarm at 1 %	5×1 = 5 Medium

Recommendation

Add a second loop oxygen analyser (2oo2) for the XV-105 permissive and record the O2 readings in the purge procedure.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R27

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R27**
Node(s)	N8 Hydrogen loop	Action party	Mechanical
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N8.9	Flow REVERSE: Gas flows back through K-101	K-101 stops with the discharge at 32.0 bar g	Reverse rotation, seal and bearing damage, hydrogen release at the seal.	None shown (no non-return valve on the P&ID)	3×3 = 9 Medium
N8.10	Temperature MORE: K-101 discharge hot	K-101 running on kickback with FV-103 closed; hot V-104 gas (see N7)	Compressor valves and seals damaged; hydrogen release.	K-101 low forward-flow alarm	3×3 = 9 Medium

Recommendation

Fit a non-return valve on the K-101 discharge and confirm the kickback / anti-surge protection and its high-temperature alarm.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R28

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R28**
Node(s)	N9 Nitrogen, vent, EDV-101 and flare	Action party	Process / Mechanical
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N9.2	Flow REVERSE: Hydrogen into the nitrogen header	HV-120 opened or passing while the loop is above the N2 header pressure (8 bar g)	Hydrogen in the plant nitrogen network: fire where the nitrogen is used for inerting.	Nitrogen only flows while the loop is below the header pressure (procedure)	4×3 = 12 High

Recommendation

Protect the nitrogen connection: non-return valve, double block and bleed or a removable spool; HV-120 opened only below the header pressure.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R29

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R29**
Node(s)	N9 Nitrogen, vent, EDV-101 and flare	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	10 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N9.3	Flow NO / NOT: EDV-101 does not open on demand	Actuator or solenoid failure; isolation valve left closed	Loop cannot be depressured in a fire or a leak: vessels weaken under fire exposure while at pressure.	HV-121 vent to flare (manual)	5×2 = 10 High
N9.4	Flow MORE: Vent and depressuring together exceed the flare capacity	HV-121 and EDV-101 open while other units relieve	Flare header back-pressure: relief from other equipment restricted.	None on the P&ID	3×2 = 6 Medium
N9.7	Pressure LESS: Depressuring too fast	EDV-101 and HV-121 open with the reactors hot	Hot liquid flashes: level swell, carry-over to E-103, R-102 and the flare; reactor contents boil.	Any SIS trip stops benzene and hydrogen before depressuring	3×3 = 9 Medium
N9.8	Temperature LESS: Metal chilled by flashing liquid during depressuring	Liquid auto-refrigerates as the pressure falls (cold, liquid-full equipment)	Carbon steel below its minimum design metal temperature: brittle fracture on repressuring.	None on the P&ID	4×1 = 4 Low

Recommendation

Depressuring study for EDV-101 (rate, flare load, minimum metal temperature); fail-open actuator and proof test.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R30

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R30**
Node(s)	N9 Nitrogen, vent, EDV-101 and flare	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	8 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N9.5	Flow AS WELL AS: Liquid to the flare	Depressuring with R-101 or V-104 level high; liquid swell on flashing	Burning liquid rain from the flare tip.	LIC-101 HI alarm at 82 %; LIC-104 HI alarm at 78 %	4×2 = 8 Medium
N9.6	Composition AS WELL AS: Air in the flare header	Flare purge lost; header opened during maintenance	Flashback and explosion in the flare header when hydrogen is vented.	None on the P&ID	4×2 = 8 Medium

Recommendation

Provide a flare knock-out drum and a continuous flare purge or seal to keep air out.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R31

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R31**
Node(s)	N1 Benzene feed, N2 R-101 slurry reactor, N6 Transfer and catalyst filter, N10 Procedures	Action party	Operations / Safety
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.7	Composition OTHER THAN: Sulphur compounds (thiophene) in the benzene	Off-spec benzene cargo; TK-100 contaminated; wrong tank lined up	The Raney-Ni in R-101 is poisoned irreversibly: conversion falls, benzene slips to R-102 and R-103 (more heat there), product off-spec; the catalyst must be replaced.	AI-103 HI alarm at 10 ppm; TI-107 HI alarm at 185 °C; fresh catalyst through the catalyst pot HV-117	3×3 = 9 Medium
N2.9	Reaction MORE: More catalyst activity than intended	Catalyst pot HV-117 left open (catalyst added continuously); fresh catalyst more active than the aged charge	Reaction rises faster than TIC-101 can follow: temperature overshoot, more isomer.	TIC-101 cascade; TIC-101 HI alarm at 208 °C; TSHH-100 trips at 220 °C	3×2 = 6 Medium
N6.7	Maintenance OTHER THAN: Filter opened for cleaning while not isolated or not gas-free	Change-over done with leaking isolation; hydrogen and hydrocarbon not purged; filter cake allowed to dry	Hydrogen and hot hydrocarbon released when the filter is opened; dry Raney-Ni cake ignites in air.	F-102 A/B selection; nitrogen through HV-120 for purging	4×3 = 12 High
N10.5	Sequence AFTER: Catalyst charged after the feed is at full rate	Catalyst top-up through HV-117 while reacting at design rate	Activity step raises the heat release: temperature overshoot.	TIC-101 cascade; TIC-101 HI alarm at 208 °C	3×2 = 6 Medium
N10.6	Maintenance OTHER THAN: Dry Raney-Ni exposed to air during catalyst handling	Catalyst allowed to dry in the pot, a drum or the filter cake	Pyrophoric catalyst ignites: fire, burns, ignition of hydrocarbon nearby.	Catalyst pot HV-117 charging under liquid (procedure)	3×3 = 9 Medium

Recommendation

Write a Raney-Ni handling procedure: always wet and under nitrogen, metered charges through HV-117, filter-cake handling.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R32

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R32**
Node(s)	N2 R-101 slurry reactor, N3 Pump-around, E-101 and steam drum, N10 Procedures	Action party	Operations / Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.2	Temperature LESS: R-101 below the light-off temperature with feed in	Over-cooling (FV-104 fully open, which is its fail position); catalyst poisoned or aged; hydrogen flow low; agitation reduced	The reaction slows while benzene keeps coming: unreacted benzene accumulates in R-101 and slips to R-102. When the cause is removed the accumulated benzene reacts at once and the temperature can jump past the trip.	TIC-101 LO alarm at 190 °C; XV-101 alarm when benzene is permitted with R-101 below 180 °C; feed permissive TI-100 ≥ 140 °C (start only)	4×3 = 12 High
N3.2	Flow MORE: Pump-around flow at maximum	FV-104 fails open (loss of instrument air) or FIC-104 output high; TIC-101 measurement reads high	R-101 over-cooled below light-off: benzene accumulates (see N2); D-101 makes more steam and its pressure rises.	TIC-101 LO alarm at 190 °C; XV-101 alarm with R-101 below 180 °C; PIC-110 HI alarm at 5.6 bar g	3×3 = 9 Medium
N10.3	Time EARLY: Feed ramped up too fast after light-off	Benzene raised to full rate before the pump-around and condenser carry the load	Heat release outruns heat removal: TIC-101 overshoots toward the trip.	TIC-101 HI alarm at 208 °C; TSHH-100 trips at 220 °C; XV-101 alarm with R-101 below 180 °C	4×3 = 12 High
N10.8	Sequence REVERSE: Shutdown steps in reverse order: hydrogen stopped before the benzene	K-101 stopped or FV-103 closed while XV-101 is still open	Benzene left unreacted in R-101 and R-102 with catalyst; heat released at the next start.	XS-102 trips the unit when K-101 stops with XV-101 open (10 s); FIC-103 LO alarm at 35000 Nm³/h	3×2 = 6 Medium

Recommendation

Light-off and shutdown steps: limit the feed ramp after XV-101 opens, stop benzene before hydrogen, and add a low-temperature-with-feed interlock (TI-100 with XV-101 open) to stop benzene accumulation.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R33

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R33**
Node(s)	N2 R-101 slurry reactor	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N2.1	Temperature MORE: R-101 above 200.0 °C toward its boiling point (about 258 °C at reactor pressure)	E-101 heat removal lost or reduced (see N3); reflux condenser lost (see N4); TIC-101 measurement reading low; TIC-101 left in MAN; too much catalyst charged; feed rate too high	Faster reaction and more isomer (MCP); at the boiling ceiling the extra heat becomes vapour, the pressure and the condenser load rise and hot hydrocarbon is carried to R-102. Overpressure of R-101 if the vapour cannot leave.	TIC-101 HI alarm at 208 °C; TI-100 independent element HI alarm at 210 °C; TSHH-100 trips at 220 °C (5 s): XV-101 and XV-105 shut, K-101 stops, E-101 to COOL with FV-104 fully open; PSHH-101 trips at 33.5 bar g	4×3 = 12 High

Recommendation

Add a TIC-101 against TI-100 deviation alarm (two elements in the same reactor).

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R34

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R34**
Node(s)	N6 Transfer and catalyst filter, N10 Procedures	Action party	Operations
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N6.2	Flow LESS: Filter F-102 restricts the flow	Catalyst fines blind the candles of the filter in service	PDI-102 rises, LV-102 opens to compensate and then R-102 level rises.	PDI-102 HI alarm at 0.8 bar; standby filter selected by F-102 A/B	2×4 = 8 Medium
N6.7	Maintenance OTHER THAN: Filter opened for cleaning while not isolated or not gas-free	Change-over done with leaking isolation; hydrogen and hydrocarbon not purged; filter cake allowed to dry	Hydrogen and hot hydrocarbon released when the filter is opened; dry Raney-Ni cake ignites in air.	F-102 A/B selection; nitrogen through HV-120 for purging	4×3 = 12 High
N10.7	Sequence PART OF: Shutdown purge incomplete before equipment is opened	Gas-freeing steps skipped or not verified (hydrogen, cyclohexane left inside)	Flammable release or fire when the flange is broken; pyrophoric catalyst exposed to air.	AI-100 and AI-105 cover the loop only, not the item opened	4×3 = 12 High
N10.10	Time LATE: Filter change-over late	PDI-102 alarm not acted on	R-102 level rises toward the gas outlet (see N5).	PDI-102 HI alarm at 0.8 bar; LIC-102 HI alarm at 80 %	2×3 = 6 Medium

Recommendation

Gas-free test and positive isolation before opening any equipment (filters, reactors); written filter change-over procedure.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R35

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R35**
Node(s)	N1 Benzene feed, N6 Transfer and catalyst filter	Action party	Mechanical
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.1	Flow NO / NOT: No benzene flow to R-101	Running feed pump P-100A stops (motor trip) and P-100B is not started; FV-101 closes (fails closed on loss of instrument air); XV-101 closed by the SIS or spuriously; TK-100 empty or suction strainer blocked	Reaction stops: R-101 cools, hydrogen consumption falls and the loop pressure rises until PIC-105 closes PV-105; production lost and a new light-off needed. A pump running against a closed FV-101 or XV-101 overheats and its seal can leak benzene.	FIC-101 LO alarm at 25 t/h; standby pump P-100B; TIC-101 LO alarm at 190 °C; PIC-105 holds loop pressure by closing PV-105	2×3 = 6 Medium
N1.5	Pressure MORE: Feed line at pump shut-off pressure (dead-headed pump)	XV-101 closed by the SIS or FV-101 closed while P-100A runs; a manual valve downstream left shut	Pump runs at shut-off: liquid heats, seal and bearing damage, benzene leak at the seal (flammable, carcinogenic).	FIC-101 LO alarm at 25 t/h; trip procedure stops the feed pump	3×3 = 9 Medium
N6.5	Pressure MORE: Pump discharge at shut-off pressure	LV-102 closed with P-102A running	Pump overheating and seal leak of hot cyclohexane with hydrogen.	LIC-102 HI alarm at 80 %	2×3 = 6 Medium

Recommendation

Fit minimum-flow bypasses on P-100A/B and P-102A/B.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R36

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R36**
Node(s)	N7 R-103, E-106, V-104 and product	Action party	Process
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	9 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N7.8	Pressure MORE: Storage tank pressure raised by dissolved hydrogen	Hydrogen dissolved at 29.0 bar g flashes when the product reaches the atmospheric tank	Hydrogen in the tank vapour space: flammable atmosphere, vent load.	None on the P&ID	3×3 = 9 Medium

Recommendation

Review the blanketing and venting of TK-106 / TK-107 for hydrogen flashing from the product.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R37

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R37**
Node(s)	N1 Benzene feed, N7 R-103, E-106, V-104 and product	Action party	Instrument
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	12 · High

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N1.2	Flow MORE: More benzene than the hydrogen and heat removal can take	FIC-101 output driven high (controller or positioner fault, set-point entry error); FI-101 transmitter reading low; FV-101 trim damaged	Conversion in R-101 falls and unreacted benzene passes to R-102 and R-103, which then release its heat of reaction: R-102 and the adiabatic R-103 bed heat up (about 9 °C per 1 % benzene slip), the product goes off-spec and the bed can run away.	FIC-101 HI alarm at 42 t/h; TIC-102 HI alarm at 172 °C; TI-107 HI alarm at 185 °C; TSHH-107 trips at 220 °C (5 s); TSHH-100 trips at 220 °C (5 s); AI-103 HI alarm at 10 ppm	4×3 = 12 High
N7.1	Temperature MORE: R-103 bed hotter than design (hot spot)	Benzene slip from R-102 (each 1 % benzene gives about 9 °C adiabatic rise); hot inlet from R-102	Bed runaway: catalyst sintered, isomer formed, vessel wall above design temperature, loss of containment.	TI-107 HI alarm at 185 °C (hottest of five); TSHH-107 trips at 220 °C (5 s); AI-104 HI alarm at 200 ppm	4×3 = 12 High
N7.9	Reaction LESS: R-103 finishing activity low	Finishing catalyst deactivated or poisoned	Benzene in product rises; the bed temperature rise disappears.	AI-103 HI alarm at 10 ppm; TI-107A to TI-107E profile	2×3 = 6 Medium

Recommendation

Add an R-103 bed-rise alarm (TI-107E minus TI-106) as early warning of benzene slip.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			

Action close-out sheet R38

Study	HAZOP · CHX-V1-SA-HZP-001 rev A	Action no.	**R38**
Node(s)	N5 R-102 and its jacket	Action party	Process / Mechanical
Drawings	CHX-V1-PID-001 · CHX-V1-PFD-001 · CHX-V1-IN-CE-001	Highest risk	6 · Medium

Raised from

Row	Deviation	Causes	Consequences	Safeguards	S×L
N5.5	Pressure MORE: Jacket at MP steam pressure on the CW side	HS-102 change-over leaves steam and CW connected together; TV-102 lets steam into the CW return	CW return and jacket overpressured, steam hammer in the CW header.	HS-102 selects one medium at a time	3×2 = 6 Medium

Recommendation

Positive segregation of MP steam and CW on the R-102 jacket (interlocked change-over) and jacket relief.

Response of the action party

Status: ☐ Accepted ☐ Accepted with modification ☐ Rejected (reason in the response) ☐ Closed (implemented and verified)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Unit owner			